Western Forest Products Limited

Tree Farm Licence 6

1999

Annual Report

Quatsino Tree Farm Licence 6 was granted to Western Forest Products' predecessor, British Columbia Pulp and Paper Company Limited, as Quatsino Forest Management Licence on October 25, 1950. On October 1, 1998, the Minister of Forests approved a significant change to the original Tree Farm Licence. Block 4 of TFL 25 was consolidated into TFL 6 as Block 2. This Block comprises the watersheds of Queen Charlotte Strait between Fort Rupert and Port McNeill, and portions of the Quatsino Sound watershed. Alaska Pine and Cellulose Limited, another predecessor company, was granted this Licence on May 22, 1958.

This Annual Report records accomplishments and data for TFL 6 for the current year 1999. If you have comments on this report, please contact the Chief Forester at (604) 665-6224 or email at Chiefforester@westernforest.com. For further information on WFP's Sustainable Forest Management Plan visit our website at www.westernforest.com.

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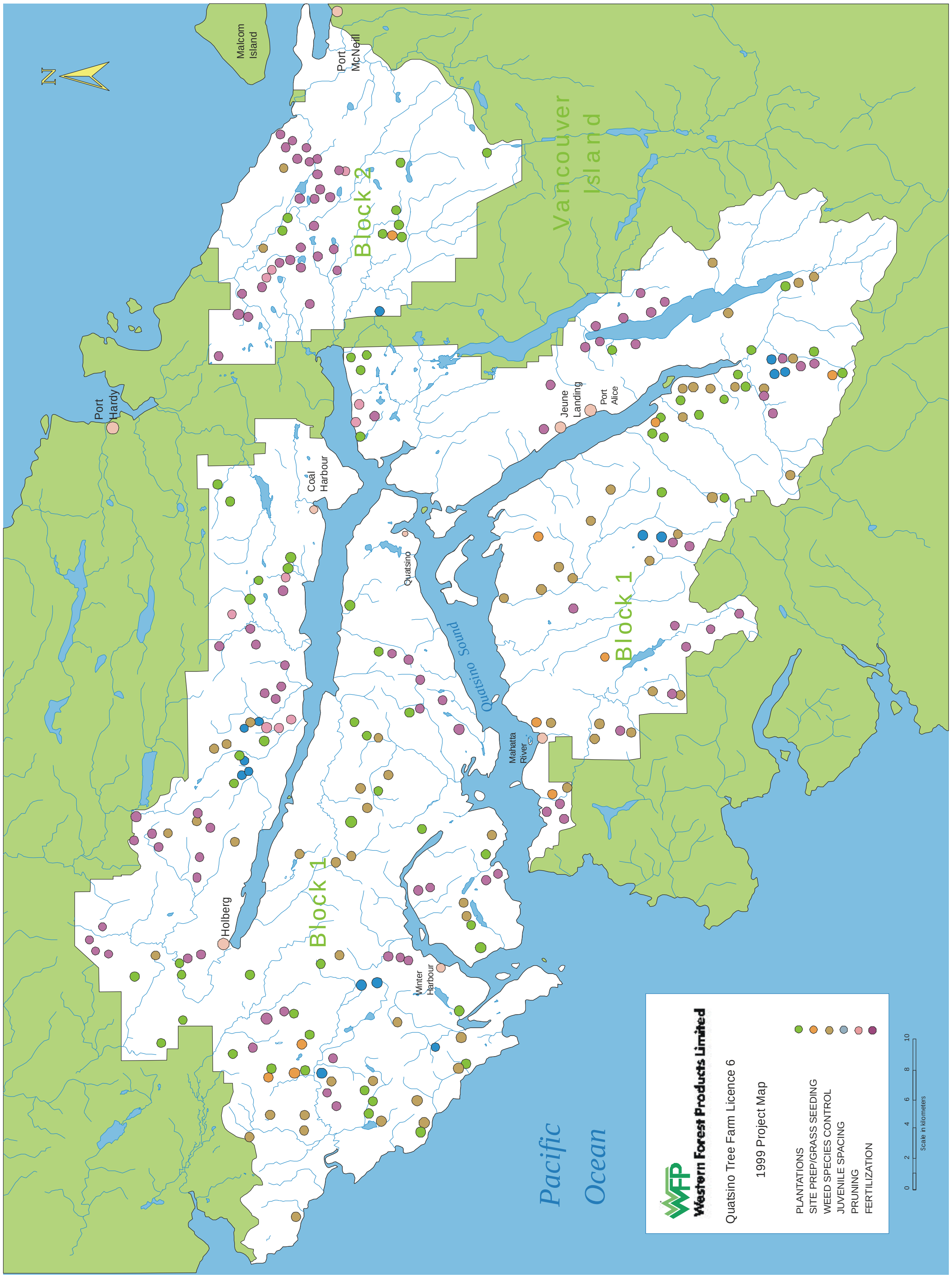
SUMMARY OF ACTIVITIES AND ACCOMPLISHMENTS – 1999

Depletion	Scaled Volume	1 375 501	m^3
	Volume Charged to Annual Allowable Cut	1 404 453	m^3
	Area Logged (Western Forest Products)	1536.3	ha
	Area Logged (SBFEP)	70.5	ha
Reforestation	Silviculture Prescriptions	1 468.9	ha
	Site Preparation	25.4	ha
	Seedlings Planted	1 088 036	trees
	Seedlings Fertilized	415 239	trees
	Area Planted	1 029.2	ha
	Stocking Surveys	1 947.2	ha
	Plantation Survival Assessments	1 447.9	Ha
	Free Growing Surveys	2 348.3	ha
Stand Management	Juvenile Spacing	317.5	ha
	Brushing and Weeding	759.7	ha
	Pruning	185.3	ha
	Fertilization	3 415.9	ha
Inventory	Cutting Permit Cruising	997	plots
	Residue Assessment	291	plots
Engineering	Roads Built	87.1	km
	Roads Re-Built	28.8	km
	Roads Maintained	815.2	km
	Roads Deactivated	53.4	km
	Roadside Seeding	41.2	km
	Site Stabilization	16.4	ha
Protection	Accidental Fires	1	
Contracting	Contractor Obligation	531 687	m^3
	Contracted	643 743	m^3
	Compliance	121	%
Minor Products	Log Salvage	1 463.5	m^3
	Shake and Shingle	11 442.3	m^3
	Cants	16.5	m^3
	Yew Bark	107.6	kg
	Gravel and Sand	26 597	m^3
	Honey	999	kg
	Cedar Oil	1227	litres
Salmon Enhancement Program (Fry Releases)	Chinook	1 133 740	fry
	Coho	708 151	fry
	Chum	40 000	fry
	Steelhead	13 243	fry
Employment	Direct Employment	229 201	person-days

PROJECT MAP


N
Malcom Island
Port McNeill
Vancouver Island
Block 2
Port Hardy
Coal Harbour
Quatsino
Quatsino Sound
Jeune Landing
Port Alice
Block 1
Block 1
Mahatta River
Holberg
Block 1
Winter Harbour
Pacific Ocean
WFP Western Forest Products Limited
Quatsino Tree Farm Licence 6
1999 Project Map
PLANTATIONS
SITE PREP/GRASS SEEDING
WEED SPECIES CONTROL
JUVENILE SPACING
PRUNING
FERTILIZATION
0 2 4 6 8 10
Scale in kilometers

1.0 INTRODUCTION

1.1 Statement of Stewardship

Quatsino Tree Farm Licence 6 encompasses 198 113 ha of coastal forest on northern Vancouver Island. It has been managed by Western Forest Products Limited and its predecessors for 49 years. More than 50 million cubic metres of logs have been produced from these lands in that time. Just less than forty percent of the total operable area of 157 000 ha remains in mature or old growth forest.

Internal and external audits and inspections indicated that a high standard of forest management was maintained during the year.

Rural Vancouver Island communities, including Winter Harbour, Quatsino, Holberg, Port Hardy, Coal Harbour, Port Alice, and Port McNeill continue to receive significant economic benefits from the management activities on TFL 6. TFL 6 provided more than 1 273 direct full-time equivalent jobs in 1999. The economic and social benefits from TFL 6, combined with the Western Forest Products environmental stewardship program have resulted in a model of coastal forest management that is always adapting and changing as new information and objectives are identified. The company continues to refine and put into practice its social, environmental and economic principles.

1.2 Statement of Sustainable Forest Management

The three important elements of sustainable forest management – social benefits, environmental appropriateness, and economic viability – are critical to the successful management of TFL 25. Key components of Western Forest Product’s sustainable forest management program include:

- Promptly reforesting all areas after harvest with ecologically appropriate species to maintain and enhance forest growth.
- Planning for the long-term in an integrated manner to incorporate the full range of forest values including soil, water, fish and wildlife, archaeology, scenic resources, and biological diversity.
- Maintaining, enhancing, and protecting forest ecosystems while providing economic, environmental, social, and cultural benefits.
- Proposing and implementing allowable annual cuts that reflect forest ecosystem capacity and sustainability as well as social and economic consideration.
- Involving the public and stakeholders in meaningful consultation of all aspects of forest management.
- Western Forest Products is actively promoting sustainable forest management and will continue to do so.

Forestry staff at the Company's Holberg Forest Operation represent many years of field experience. Mike Dietsch, Chris Nunn, Russel and Shelagh Fox, Amy Beetham, Anna Bellows (summer student) and Jeff Mosher provided excellent service in 1999 in managing WFP's largest field operation. Many staff changes occurred subsequently at Holberg in the forestry department.

1.2 Operations Highlights

Western Forest Products' parent company Doman Industries Limited had a slight improvement in its financial situation in 1999. Year end results continued to be affected by depressed demand for hemlock lumber in Asian markets, restricted access to the US markets, low pulp prices and high stumpage and logging costs. Losses for the year were $55,913,000 compared to a $74,000,000 in 1998. Sales for the year were up almost $100 million to $873 million compared to $779 million in 1998. The outlook for 2000 was expected to be significantly better with improved pulp prices and demand by mid year. Lumber sales were approximately $500 million, pulp sales $262 million and log and sawmill revenues approximately $100 million.

The Softwood Lumber Agreement with the United States continues to hamper the company's return to profitability by forcing the mills to operate at less than optimum level. The sawmill capacity for Doman Industries is 1.2 billion board feet but only 717 million board feet were produced, or 60% of capacity.

Total timber harvest for the year continued in an undercut position. A total of 3.3 million m^3 or 74% of available harvest volume was accomplished.

Stumpage payments were $84.7 million, an average of $25.28 per cubic metre. This is slightly lower than in 1998 but still a significant cost factor on an average log sales value of slightly over

$100 per cubic metre. Joint government/industry cost and administration action teams addressing stumpage and bureaucratic relief continued work in 1999. Appendix XIX summarizes the operating statistics for the Doman Group of companies for 1999.

WFP's Port McNeill Forest Operation provided a large red cedar log for an impressive new Kwakiutl totem at the provincial museum in Thunderbird Park in Victoria. This beautiful pole was carved by Johnathan Henderson and Sean Whonnock.

Company Highlights

- In an effort to address the underharvest situation in various tenures and provide new opportunities for First Nations involvement in forestry activities, the company agreed to underutilized AAC being transferred to six coastal First Nations within their traditional territories. By year-end the specific mechanism of transfer was still to be worked out in conjunction with the Ministry of Forests. It is expected the company will recover most of this volume through log purchase and cooperative development agreements.
- A total of 263 kilometres of road were constructed and 535 km were deactivated, 116 permanently. A total of 1960 km of roads were maintained in the company's 30 logging operations.
- The unresolved land use situation and lack of progress in the LRMP in the Central and North Coast Region continued to attract international attention and a market action campaign continued against companies working in the region, including Western Forest Products. By year end discussions commenced with local and international environmental groups on a new conflict-free approach to attempting to design an ecosystem based forest management model.
- The provincial government announced a major forest policy review under the direction of the Jobs and Timber Accord advocate, Gary Wouters. WFP provided a brief outlining five major areas of

much needed forestry reform for returning profitability to the coastal forest industry and sustaining environmental and social values.

- The company had an excellent year with respect to Forest Practices Code compliance with only 6 minor violations recorded out of a total of 721 inspections of company operations for a year-end 99% compliance record.

- Intensive work continued on developing the company's Environmental Management System and proceeding with work Forest Stewardship Council certification in three tenures in North Vancouver Island Region

- More than 80% of the company's timber harvesting was patch clearcutting with reserves. Average openings were less than 30 hectares in size. WFP also committed to establishing trials to assess variable retention potential in North Vancouver Island Region. Approximately 91% of the timber harvest was in mature stands older than 140 years and mostly 250 years or greater. A total of 4 146 hectares were logged and 2 915 hectares planted.

Specific TFL 6 Highlights

- The official opening of the Marble River Chinook rearing channel was held in the nearby recreation site. The Friends of Marble River Society raised more than $300,000 for this project with a design capacity to rear up to 2 million salmon fry using water drawn by a massive syphon from the Marble River.

- A record 1 994 783 salmon fry were released from the three company salmon hatcheries operating in TFL 6.

- The largest fertilization project in company history covering more than 3 400 hectares was completed in the Spring of 1999.

Forestry field monitoring staff check aerial fertilization progress during spring projects at Jeune Landing. Screened traps capture fertilizer for analysis of spread and density of application. A total of 3400 hectares were fertilized in the TFL.

- On New Years morning the Holberg Community hall burned to the ground. The facility had a previous life as a cold storage unit at the former Coal Harbour Whaling Station and was moved to Holberg in the late 1960's. By year end a new hall was constructed.
- The Forest Tour program was very successful in 1999. One hundred and fifty-eight people participated on 10 tours. Holberg's tour was voted one of the most popular by many of the participants due in part to the scenic ride down the Holberg Inlet and the of harvesting activity viewed.
- Holberg Forest Operation underwent 98 harvest inspections by the Ministry of Forests during 1999. 100% of these inspections showed that operators were in compliance with the regulations.
- A permanent 29 m steel and concrete bridge was constructed across the San Josef River at 14 km replacing an old spruce stringer bridge. Bridge replacement cost $108,000.
- The Palmerston recreation site was constructed at the end of Hecht Main, linking the road to a long stretch of west coast beaches just to the north of the Raft Cove Provincial Park.
- Jeune Landing Operation helicopter logged 8.2 ha of timber, 6.6 ha of which was blown down salvage. As well, Jeune Landing reduced the amount of road to build by Skyline logging over 20 year regeneration, without damaging second growth.
- WFP funded a full time forestry coordinator to aid the Kwakiutl First Nation in capacity building. A training program at Port McNeill saw 12 band members trained in logging. Three members of the training crew were hired full time to fill WFP logging vacancies.
- In the Port McNeill Forest Operation two major bridges were completed during the year in cooperation with Weyerhaeuser on South Hardy Main across the Coetwaas River and on Rupert Main across the Keogh River.
- Port McNeill Forest Operation also began building road and logging on the east side of Victoria Lake. Port McNeill Operation last worked in this area in the mid-1970's.
- WFP and the Quatsino First Nation co-operated in the management and harvest of the Quatsino Woodlot Licence (WL). To provide greater efficiency WFP harvested a portion of the WL volume with WFP's equipment on Quatsino's behalf while Quatsino harvested volume over and above their contract in the TFL for WFP with their tower. This allowed both parties to use more efficient equipment for mutual benefit.
- Tree of Life Essential Oils produced 1 227 litres of Western Red Cedar oil from 117 pick-up truck loads of Cw foliage collected from TFL 6 BLK 2. The project employed 6 people (5 First Nations) and has spawned a secondary business of First Nations Soap and Candle Works.

- The primary silviculture contractor for the Quatsino band closed her company in September. This resulted in the Kwakiutl and Quatsino silviculture workers forming a mixed band crew to complete silviculture work within the traditional territory of the Quatsino band. In 1999, First Nations silviculture crews completed 111 ha of spacing, 101 ha of pruning and 68 ha of brushing.
- The Kwakiutl band had a five-week silviculture-training program. The Company provided the pruning, spacing and creek cleaning areas for the field portion of the program. WFP also had a two-month intern position for one band member to do silviculture surveys and layout alongside our summer student.

A Holberg camp Christmas party in the 1940's at which Mr. Humphries, Camp Manager for B.C. Pulp Ltd., addresses the logging crew and spouses during the festivities. At that time, Holberg was one of the largest float based logging operations in the world. Note the pastries and pinstripe suits!

2.0 MANAGEMENT AND OBLIGATION PERFORMANCE

2.1 Volume

Scaled timber production for the year was 1 375 501 m^3. Volumes by timber mark and Operation are summarized in Appendix I. The volume charged to the allowable cut including scaled production and residue survey volumes was 1 404 453 m^3 (Appendix II).

Log scaling and marking staff, Lee-Ann Watson and Brian Thompson, at the Quatsino Dryland Log Sort are part of a team which scales and processes more than 800,000 cubic metres of logs each year at this central facility for TFL 6 and other company tenures on Quatsino Sound.

2.2 Area

The Company harvested timber from 1 536.3 ha during the year. Appendix III summarizes the denudation by Operation. Twelve per cent of the area logged was classified as medium, low, or very low ecosites. The low site partitioned AAC volume requirement was significantly exceeded (Appendix IX). The non-productive area in the TFL was increased by 87.1 hectares to account for the formerly productive area occupied by new roads.

2.3 Contractor Compliance

Harvesting and forest management activities on TFL 6 during the year employed contractors and Company personnel. The minimum Contractor Clause commitment was exceeded by 21 per cent (Appendix VI). Appendix VII lists the phase and full-harvesting contractors. Appendix VIII lists all contractors employed in harvesting, silviculture, and resource management on the TFL.

2.4 Planning

2.4.1 Land and Resource Management Plans

During the year little progress was made by government on completing the Vancouver Island Land Use Plan. In anticipation of VILUP declaration by cabinet, WFP continued special projects in the TFL's 2 Special Management Zones (SMZ). These include enhancing the retention of stand structure in and adjacent to SMZ harvest blocks, the addition of a new recreation site to SMZ 2, continuation of a Resource Inventory Committee (RIC) standard fisheries inventory project for SMZ 2 and 4, and a wildlife habitat modelling project centred on the two SMZs.

Forestry costs associated with higher level planning, including land and resource management planning and landscape unit planning, were $129,892.

Maintaining stable trees in Riparian Reserve and Management Zones is a critical aspect of cutblock design and forest practices in TFL 6. WFP has been conducting operational trials with helicopters to reduce windfall of riparian trees using aerial pruning along the leading exposed edges of openings. Pruning technology continues to be developed.

2.4.2 Forest Development Planning

1999 was the first year that Development Plans were completely developed by the field operations. Forest Development Plans (FDP) were completed in all Operations according to Ministry of Forests' Policies and Procedures. New FDPs were approved for Holberg Forest Operation Regular Operations (January 29 – 1 year term), Holberg Forest Operation Special Management Zones (January 29 – 1 year term), Port McNeill Forest Operation (March 23 – 1 year term), and Jeune Landing (August 20 – 2 year term). Digital FDP maps were prepared by a Port McNeill mapping contractor. Forest Development Plans preparation and compilation costs totalled $87,808.

2.4.3 Silviculture and Harvest Planning

2.4.3.1 Cutting Permits and Cutting Permit Cruising

The number of active cutting permits in TFL 6 increased from 51 in 1998 to 60 in 1999 continuing the disturbing trend seen since the Ministry of Forests moved to Forest Licence styled permits for TFLs. Prior to this policy change only 9 roll-over cutting permits were required to harvest the same volume. Associated administrative costs have increased accordingly. Table 1 lists the active cutting permits, locations and expiry dates.

Cutting permit cruising plot costs decreased from 1998 costs. Due to a high proportion of plots in lower cost operations and the contractors' continued co-operation in a Company-wide cost reduction program. Average cost per established plot for TFL 6 was $142. Average cost per compiled plot was $59. The weighted plot cost was $84. A total of 958 cutting permit cruising plots were established in TFL 6. Twenty-two cruise reports were compiled using data from 2 297 plots.

Total contractor cost was $136,145. Forestry department costs associated with cutting permits and cutting permit cruising including contractor costs totalled $263,572.

Table 1 Active Cutting Permits

CP	Expiry Date	Location
103	Apr 30/00	Holberg
104	Jul 31/00	Holberg
105	Dec 31/00	Holberg
200	Mar 31/00	Jeune Landing
201	Mar 31/00	Jeune Landing
202	May 31/00	Jeune Landing
203	Oct 31/99	Jeune Landing
204	Mar 31/00	Jeune Landing
205	Aug 31/00	Jeune Landing
206	Oct 13/01	Jeune Landing
207	Aug 10/01	Jeune Landing
208	Mar 31/01	Jeune Landing
210	Nov 17/01	Jeune Landing
300	Mar 31/00	Botel
301	Jan 31/00	Botel
302	Feb 29/00	Botel
400	Mar 31/00	Mahatta River
401	Aug 31/99	Mahatta River
402	Aug 31/00	Mahatta River
403	Sep 30/00	Mahatta River
404	Jan 31/01	Mahatta River
405	Sept 23/01	Mahatta River
406	Jul 27/01	Mahatta River
407	Dec 5/01	Mahatta River
4M	Aug 31/00	Port McNeill
40M	Jan 01/01	Port McNeill
43M	Oct 31/00	Port McNeill
44M	Dec 31/00	Port McNeill
500	Nov 30/00	Port McNeill
501	Apr 30/01	Port McNeill
502	Apr 30/01	Port McNeill

CP	Expiry Date	Location
503	Apr 30/01	Port McNeill
504	Apr 30/01	Port McNeill
505	Aug 12/01	Port McNeill
601	Mar 31/00	Port McNeill
602	Feb 29/00	Port McNeill
603	Jun 30/00	Port McNeill
604	Jun 30/00	Port McNeill
605	Jul 5/00	Port McNeill
701	Feb 29/00	Winter Harbour
702	Feb 28/00	Winter Harbour
703	Dec 31/00	Winter Harbour
704	Oct 21/00	Winter Harbour
705	Dec 14/00	Winter Harbour
750	Jul 31/00	Holberg
751	Aug 31/00	Holberg
752	Nov 30/00	Holberg
753	Dec 14/01	Holberg
801	Feb 29/00	Koprino
803	Jan 31/00	Koprino
805	Jul 31/00	Koprino
872	Dec 31/00	Michelson
873	Jul 31/00	Michelson
874	Jul 8/00	Michelson
900	Jan 31/00	Port McNeill
901	Jan 31/00	Port McNeill
902	May 31/00	Port McNeill
90	Oct. 31/01	Holberg
97	Apr 30/00	Jeune Landing
98	Apr 14/00	Port McNeill

2.4.3.2 Road Permits

The Company maintained three Road Permits in 1999, as listed in Table 2.

Table 2 Active Road Permits

Road Permit	Expiry Date	Operation
R06936	Open	Port McNeill
R06897	Open	Jeune Landing and Mahatta River
R06896	Open	Holberg, Botel, and Winter Harbour

2.4.3.3 Silviculture Prescriptions

Forty-seven new silviculture prescriptions were submitted during the year (Table 3). The total area for which new silviculture prescriptions were submitted was 1 468.9 ha. Average block size including reserves was 29.9 ha. Average reserve area varied from 10.9 per cent to 11.8 per cent by Operation, with a weighted average of 11.4 per cent.

Forestry costs for new silviculture prescriptions and amendments to previously submitted prescriptions totalled $130,044 or $89 per ha.

Table 3 Silviculture Prescriptions

Operation	Submitted SPs (#)	Total Area (ha)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)	Average Reserve (%)
Holberg	16	579.0	36.2	6.8	50.0	11.8
Jeune Landing	20	665.0	33.0	13.0	44.0	11.3
Port McNeill	11	224.9	20.4	1.1	43.2	10.9
TOTAL	47	1468.9	29.9	1.1	50.4	11.3

2.4.4 Public Involvement

2.4.4.1 Forest Development Plans

Four new plans for Port McNeill, Jeune Landing, and Holberg (2) were approved in 1999. All plans were presented at community open houses as well as being available for sixty days at WFP and Ministry of Forests offices. Three of the open house sessions occurred in calendar 1998 and are not recorded in this Annual Report.

A summary of the responses from the public is detailed in Table 4. Special presentations were also made to the Kwakiutl and the Quatsino First Nations to explain activities within their traditional territories.

Table 4 Forest Development Plan Reviews

Operation	Plan	Location	Attendance	Written Responses	Oral Responses
Jeune Landing	Jeune Landing FDP	Port Alice	10	3	0

2.4.4.2 Pest Management Plans

Jeune Landing Operation submitted the first TFL 6 Pest Management Plan. The operation held public review meetings for Pest Mangement Plan 2000-2004. Advertisements were placed in the local paper and letters of notification were sent to the Village of Port Alice, the Colonial Hatchery and Victoria Lake residents. Five Victoria Lake residents responded. A meeting and subsequent discussion was also held with the Quatsino Band.

2.4.4.3 High Conservation Value Forest

Western Forest Products met with 404 people through 22 separate meetings to present Western Forest Products' Forest Stewardship Council Certification initiatives. The High Conservation Value Forest Public Statement was presented at these meetings and was sent out to 250 stakeholders including First Nations and environmental organizations. The statement was also located on Western Forest Product's website. Several written responses to the statement were received. Consultation will continue on High Conservation Value Forests and the statement will be updated to incorporate feedback.

2.4.4.4 Forest Education and Special Events

The Company continued its support of the North Island Forestry Centre (NIFC). This non-profit society co-ordinated summer tours of logging operations on the North Island. Operations hosted a total of 314 guests on 25 tours. These tours included forest management activities and discussions on forest practices with Western Forest Products employees. Holberg Operation also conducted a special tour for 24 students from a New York City high school. All Operations made presentations to elementary school students for National Forestry Week. As well, the Port McNeill Operation hosted a special tour for a group of school students involved with "Reach for the Treetops". Many company foresters participated in "Career Day", "Interview Day" and other School District activities.

North Vancouver Island Region hosted numerous special tours. The international board and executive of Forest Stewardship Council toured the North Island and discussed WFP's certification initiatives with keen interest. David Ford, President of Certified Forest Products Council, also toured the North Vancouver Island Region with an interest in certification. Several tours were also hosted for customers wanting to see WFP's forest management first hand.

The Company spent $75,975 on tours and special events in TFL 6. In addition, the Company's Public Projects fund contributed $14,566 to Forest Education efforts in the TFL. Additional public relations costs associated with Forest Renewal BC events or projects are covered by that Multi-year Agreement.

WFP continues to host tours of its North Vancouver Island Region forest operations throughout the summer. The North Island Discovery Centre is supported by major forest companies. Each WFP operation organizes weekly tours including stops such as the Port McNeill Dryland Sort with old growth logs as an impressive backdrop.

2.5 Inventories and Mapping

2.5.1 Geographic Information System and Electronic Mapping

Five GIS stations were used for corporate resource inventory mapping and analysis. WFP continues to use Pamap as its GIS system of choice; however the use of ArcView, Microstation GeoGraphics, FME and World Construction Set are being utilized to perform a multitude of analysis and data processing tasks.

Microstation (CAD) systems are utilized for operational mapping. Currently, the North Vancouver Island Region has been using a contractor for mapping work. The integration of digital data to and from the mapping contractor has greatly improved efficiencies within the mapping department.

Training opportunities and support provided by PCI Geomatics, Pacific Alliance Technologies and BCIT were used to advance the skills of GIS personnel.

Prorated TFL 6 GIS costs totaled $54,988.

2.5.2 Forest Inventory

TFL 6 digital files were updated to January 1, 2000 to incorporate changes to forest cover, roads and logging history that occurred during 1997 – 1999. Silviculture history was restructured within the GIS to allow each silviculture activity to be stored separately. This facilitates easier tracking of the different activities on-going in the TFL and removes the need to generalize the data.

The shifting of the TFL digital mapping to the BCGS UTM NAD83 datum has been completed.

Photo interpretation and quick plot ground sampling for the new Vegetation Resource Inventory (VEG) started in TFL 6. This new inventory will provide planning and field staff with improved information regarding second growth forest.

Cost for forest inventory and operational mapping work totaled $221,098.

2.5.3 Ecosystem

Minor ecosystem classification revision work was undertaken in the TFL. Costs were $6,388.

2.6.4 Terrain Stability

Minor terrain stability classification revision work was undertaken in the TFL. Costs for terrain stability work totaled $3,490.

2.6.5 Geotechnical Analyses

Engineering consultants were employed to complete geotechnical analyses of cutblocks and road construction and deactivation projects throughout TFL 6. Table 5 details the consultants employed by Operation. Costs for geotechnical analyses are included in logging costs.

Table 5 Geotechnical Analyses

Operation	Geotechnical Firm	Cutblocks	Roads
Jeune Landing	EBA Engineering Consultants	6	
	Westrek Geotechnical Services	5	
	Arbor Tek	5	
Port McNeill	EBA Engineering Consultants	2	
	Arbor Tech	1	
Holberg	Westrek Geotechnical Services	20	2
TOTAL		39	2

2.6.6 Resource Inventories and Assessments

A large number of inventories and assessments were competed during the year, many to support the preparation of operation plans. Except where indicated, costs for these inventories and assessments were incurred by the Forestry Department, and totalled $229,935.

2.6.6.1 Archaeology

Sources Archaeological & Heritage Consultants completed 7 Archaeological Impact Assessments (AIAs) in cut blocks in the Holberg, Winter Harbour, Koprino, Port McNeill and Coal Harbour operating areas. Representatives of the Quatsino First Nation were involved in all of these assessments. Operational plans were adjusted to protect Culturally Modified Trees and other archaeological features identified in these assessments. Archaeological Impact Assessment reports were provided to the Quatsino First Nation, the Ministry of Forests and the Heritage Conservation Branch. Costs for cutblock archaeological assessments are included in logging costs.

2.6.6.2 Streams

Stream classification assessments were carried out on 37 cutblocks and roads during the year in advance of final cutblock and road engineering. This information was used to designate riparian reserves and riparian management zones in each cutblock. Fishfor Contracting completed all surveys. Costs of stream classifications are included in logging costs.

Western Forest Products continued with the fisheries inventory work in the Special Management Zone. Work was done to Resource Inventory Committee (RIC) standards.

Forestry department costs for stream assessments and inventory work totalled $120,483. Reimbursements from FRBC for landscape level stream work was $111,245.

A special riparian maintenance project was undertaken on the Keogh River near Port McNeill to create "old growth" conditions supportive of the stream ecology. Bob Toth of Sigfor Services undertakes conifer girdling to provide large organic debris for eventual entry into the stream. Vince Poulin, Fisheries Consultant, has been developing sophisticated prescriptions to sustain riparian habitats.

2.6.6.3 Wildlife

Wildlife work in TFL 6 focused on developing and testing a number of modeling algorithms designed to predict suitable habitat for a variety of species. Using attributes from resource inventories (i.e. forest cover, topography, ecosystem), GIS and database routines were developed to predict habitat suitability for Marbled Murrelet, Deer, Elk and Black Bear. The algorithms used in this project were developed based on current research work ongoing in the province. Fieldwork was also completed as part of this project to test the accuracy of the models. Overall, this project yielded good results and has provided planning staff with a useful tool for completing strategic and operational plans. Costs for this work totaled $100,286.

2.6.6.4 Recreation

The TFL 6 Recreation Inventory was reviewed during the year. As there have been no significant changes in recreational use of TFL lands it was not considered necessary to revise the inventory. Costs totaled $3,872.

2.6.6.5 Landscape Assessments

The majority of work completed throughout the year focused on completing the Regional Landscape Unit Planning Strategy (RLUPS). The original RLUPS, completed in 1997, included the delineation of all landscape unit boundaries, the determination of biodiversity emphasis (lower, intermediate and higher) for each landscape unit and the scope and priorities then proposed for LU planning. The work complete in 1999 consisted of the review and update of the 1997 RLUPS, which resulted in a set of revised LU planning strategies. These strategies include the following information:

- A statement confirming the satisfactory completion of the review and compliance with the Chief Forester's Higher Level Plans: Policy and Procedures (HLP:PP);
- A matrix confirming that each element of Chapter 5, HLP:PP has been reviewed;
- A summary of red flagged units and how these were addressed or will be addressed in the review;
- A tabular summary of landscape units showing biodiversity emphasis options area, and a proposed priority based schedule to legally establish LUs and priority biodiversity objectives, within three years.
- A summary of major changes and issues dealt with in the review and a description of the reason for the changes.

The next step in continuing landscape unit planning is for the development of *priority biodiversity* objectives within a three-year time period. Priority biodiversity planning is defined by the Landscape Unit Planning Guide to consist of the retention of old growth forest; and stand structure through wildlife tree retention.

General information regarding the landscape units found within TFL 6 and the proposed timeline to complete the priority biodiversity objective follows:

Table 6: Landscape Units of TFL 6

Landscape Unit	BEO	Gross LU Area (ha)	LU Area (ha) within TFL	Legal Establishment	Priority
Marble	Intermediate	60,440	23,649	Dec 2000	High
San Josef	Intermediate	90,060	61,607	Dec 2000	High
Mahatta	Low	43,148	24,440	Dec 2000	Medium
Keogh	Low	58,288	31,337	Dec 2000	Medium
Neroutsos	Low	26,523	26,523	Dec 2000	Low
Holberg	Low	35,582	33,500	Dec 2000	Low

The bald eagle *(Haliacetus leucocephalus) is an important wildlife species within TFL 6. WFP maintains a comprehensive eagle nest inventory involving more than 100 nests in the TFL. Specific management practices are implemented when operating around eagle nests.*

3.0 MANAGEMENT OBJECTIVE ACHIEVEMENTS

3.1 Management and Utilization of Timber Resources

3.1.1 Harvesting Methods

Six logging systems were used in TFL 6 in 1999 (Figure 1). In areas managed by Jeune Landing (including Mahatta River) grapple yarders harvested the largest portion of the volume, 47 per cent, followed by the supersnorkel, 23 per cent, and the tower, 18 per cent. Both hoe forwarding and skyline logging systems accounted for 5 per cent of the volume harvested at Jeune Landing. Jeune Landing was the only operation to use the heli-logging harvesting system and produced 2 per cent of the total volume by this method.

1999 was the second year that Jeune Landing harvested with the skyline logging system. The 046 Skadill produced between 270 m^3 and 300 m^3 per day. The amount of road to be built was reduced by skyline logging over 20 year old regeneration. The skyline system caused no damage to the regeneration. Jeune Landing heli-logged a total of 8.2 hectares of which 6.6 hectares was blowdown salvage.

Holberg, Botel, and Winter Harbour used grapple yarder, super snorkel, and hoe forwarding logging systems (Figure 1). Grapple yarders harvested 41 per cent of the volume, hoe forwarders 35 per cent, and super snorkels 24 per cent.

Port McNeill Forest Operation used hoe forwarders, a tower, a super snorkel, and grapple yarders (Figure 1). The super snorkel handled 33 per cent of the volume. Grapple yarding accounted for 28 per cent of the volume, and hoe forwarding accounted for 27 per cent of the volume. The entire tower logging volume, 12 per cent, was completed by Quatsino Forestry.

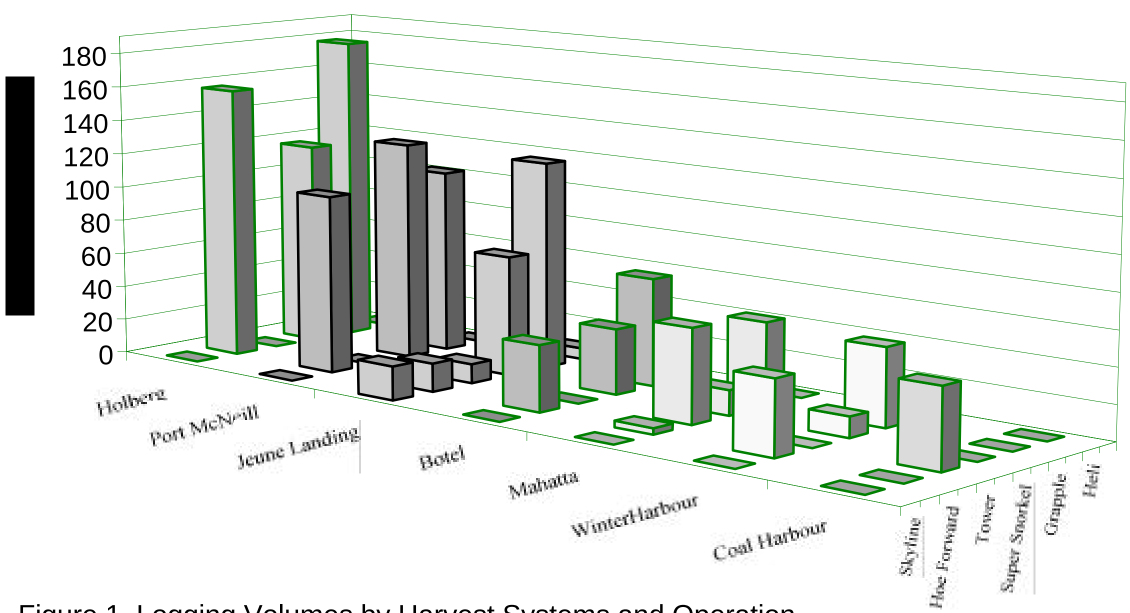


Figure 1 Logging Volumes by Harvest Systems and Operation

Jeune Landing loggers pose with a grapple from the company yarder. Grapple yarders are used to log almost 40% of annual production in TFL 6 and produce an average of 250 m³ per work day.

3.1.2 Silviculture Systems

Patch clearcut with reserves is the main silvicultural system employed at all Operations. Fifty-eight of the 60 areas harvested in 1999 were clearcut with reserves. All of the sixty-four blocks engineered during the year will use this system. Average reserve area exceeds ten per cent of the gross area of each block. Table 7 provides a silviculture system summary by Operation.

Table 7 Silvicultural Systems by Operation

Operation	System	Blocks Engineered (#)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)
Holberg	Clearcut with Reserve	26	38.95	6.8	58.4
Jeune Landing	Clearcut with Reserve	15	34.3	19.0	44.8
Port McNeill	Clearcut with Reserve	23	18.9	0.16	46.2

Operation	System	Blocks Harvested (#)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)
Holberg	Clearcut with Reserve	25	35.9	8.4	52.9
	Clearcut	1	6.3	6.3	6.3
Jeune Landing	Clearcut with Reserve	15	32.3	19.5	44.4
	Clearcut	1	13.9	13.9	13.9
Port McNeill	Clearcut with Reserve	18	26.6	2.3	55

3.1.3 Felling, Bucking, and Utilization Specifications

3.1.3.1 Specifications

Cutting permits outline the required and optional utilization standards. In general:

i. Maximum stump height of 30 cm on shortest side;

ii. All old growth coniferous trees containing X grade logs or better will be utilized to a top diameter of 15 cm inside bark (10 cm for second growth);

iii. All conifer trees or parts of these trees exceeding 3 m in length which contain X grade logs or better will be removed;

iv. Logs or parts of logs less than 3 m in length and broken at ends are classed as breakage;

v. Logs will not be bucked or trimmed in a manner that reduces grade.

These requirements apply to all living and dead trees that meet the standards. Quality control is maintained by WFP staff and monitored through felled and bucked inspections and residue surveys.

Mike Murray, a contract faller at Jeune Landing, works in large hemlock timber in area 827. All falling in TFL 6 is undertaken by contractors. On average fallers produce about 135 m³ per day.

3.1.3.2 Residue Surveys

The Company continues to survey and report residue volumes on a calendar year basis for all cutblocks where logging is complete.

There were a total of 291 plots established during the 1999 survey year. The area surveyed in 1999 was lower than historic levels as a result of market induced production curtailments at the end of 1998 and due to policy changes and issues of survey procedure. For these areas surveyed in 1999, AAC depletion totaled 43 011.6 m^3 or 38.0 m^3/ha, while the billable waste volume totaled 12 310.5 m^3 or 10.9 m^3/ha (Table 8).

The contractor costs for residue sampling and compilation were $26,635 or $91 per plot. Total costs of residue assessments were $56,957.

Table 8 Residue Assessment Survey Summary

LOCATION	SURVEY AREA		AAC DEPLETION		BILLABLE WASTE TOTAL VOLUME		NUMBER OF PLOTS				
	Gross (ha)	Net* (ha)	m^3/ha	m^3	m^3/ha	m^3	Slash	Road Side	Pile	Other	TOTAL
Holberg	233.8	221.2	43.3	9570.5	10.0	2202.9	25	18	18	4	65
Winter Harbour	203.7	196.3	45.7	8980.1	10.2	2004.4	16	16	14	4	50
Koprino	156.5	148.0	42.6	6308.6	12.6	1871.8	10	10	10	0	30
Botel Lake	34.4	33.2	18.8	623.3	6.5	214.8	4	4	4	0	12
Port McNeill	169.7	162.1	36.6	5928.5	7.4	1195.8	11	10	11	4	36
Coal Harbour	107.4	99.6	25.2	2513.2	10.8	1071.3	10	10	11	4	35
Jeune Landing	147.5	141.4	35.4	4999.2	15.4	2178.6	10	10	10	4	34
Mahatta	134.7	129.4	31.6	4087.5	6.2	808.3	11	8	8	2	29
TOTAL	1187.7	1131.2	38.0	43010.9	10.9	11547.9	97	86	86	22	291

3.1.3.3 Special Forest Products

Special Forest Products harvested in 1999 included salvage logs, cedar shake and shingle blocks, cedar cants, yew bark, honey, cedar oil, and gravel (Appendix XI). Salal was harvested in the TFL, however, amounts were not reported or recorded.

More than 11 000 m^3 of shake and shingle blocks were salvaged (Appendix XI). Jeune Landing reported similar amounts to the previous year's salvage at 2 179 m^3. Both Port McNeill and Holberg reported lower (approximately 10 per cent) salvage volumes for shake and shingle than the previous year. Holberg also moved into the Forest Licence to salvage. Port McNeill reported 4 701.2 m^3 and Holberg reported salvage of 4 562.1 m^3.

Yew bark harvesting continued at Port McNeill although the total weight of the 1999 harvest, 107.6 kg, was significantly lower than last year's harvest of 2 932 kg (Appendix XI). In 1999 no yew bark harvest was reported by the Jeune Landing Operation compared to 1998 when 1 352.0

kg were harvested. Yew bark harvest was lower in 1999 as areas available for harvest contained no yew wood.

Port McNeill Forest Operation sold 26 597 m^3 of gravel in 1999. This represents a decrease 15 per cent decrease from the previous year.

The cedar oil contractor continued producing cedar oil at Port McNeill Operation. Processing of cedar boughs produced 1227 litres of cedar oil, 2.4 times more than last year. Honey production was greatly reduced in 1999, only 999 kg (Appendix XI) were reported, compared to 18 135 kg in 1998. One of the independent honey producers claimed 1999 as a failed year.

Staff at the Port McNeill Operation participated in a local Non-Timber Forest Products workshop. The Forestry Department administered the Minor Products program at a cost of $17,486.

3.1.4 Partitioned Harvest

The Annual Allowable Cut for TFL 6 has been partitioned into Conventional, Helicopter, and Low Site Volumes. Appendix IX reports the 1999 partitioned volume as required in Management Plan 8. The low site volume is estimated by matching low site harvest area to inventory volumes. On this basis Western Forest Products' produced 94 928 m^3 from low site lands which significantly exceeds the low site target of 52 000 m^3 per year. 6 221 m^3 were helicopter logged in the Jeune Landing Operation during the year.

3.1.5 Annual Allowable Cut and Cut Control Performance

The Allowable Cut available to Western Forest Products on TFL 6 for 1999 was 1 464 264 m^3 (Appendix IV). The chargeable cut for the year, including the scaled production and the recognized residue survey volumes, was 1 404 453 m^3 (Appendix II). The cumulative volume charged after five years of the current five-year cut control period is 6 536 720 m^3 or 95 per cent of the five-year total allowable cut (Appendix IV).

The historic cut control is presented in Appendix V. Western Forest Products and its predecessors have harvested 49 999 825 m^3 since the granting of the Licence in 1950. To the end of 1999, the accumulated Allowable Cut was 49 983 530 m^3. The rate of harvest is at 100.03 per cent of the Allowable Cut to date as determined by the Chief Forester of BC.

3.1.6 Small Business Forest Enterprise Program

The Small Business Forest Enterprise Program (SBFEP) has been in place since 1988. There has not been any new Block 1 SBFEP volume added to the program since a 1996 volume exchange agreement was reached with the Ministry of Forests. At the time that the agreement was made, there was still a "backlog" of TFL 6 SBFEP volume that the Ministry had not issued as Timber Sales. This backlog volume is still available to the Program. However, none of this backlog volume was harvested in 1998. An estimated 132,600 m^3 remains to be harvested in Block 1.

The SBFEP allocation in Block 2 is 19,373 m^3 per year. The Port McNeill Forest District reported under the SBFEP 36 ha in Block 2 harvested by B.W. Creative Wood Industries and 34 ha in Block 1 was harvested by Lukwa Mills of Port Hardy. The chargeable volume was 60,060 m^3 (Appendix X).

3.2 Protection and Conservation Measures

Western Forest Products protected and conserved identified resource values in TFL 6 during the year. The following sections discuss planning and operational measures taken to protect other resource values. Blocks for which amendments were submitted and blocks in which harvesting was not completed are not included in Table 9.

At the Port McNeill Forest Operation, the "batwing", a locally developed tool seen at mid stem in the picture, is used in conjunction with a grapple yarder and hoe forwarder to directionally fall timber away from stream reaches. Innovative techniques are used to address important stream protection issues. The extra costs of these activities exceed $7/m³ for the timber involved.

3.2.1 Visual Quality

Nine of the 64 blocks engineered during the year required special engineering to address visual concerns (Table 9). Two of the 60 blocks harvested during the year had been similarly modified to address visual issues. Modification includes movement of backlines and the location of reserves to minimize the visual impacts of harvesting. Areas where visual quality concerns exist were identified in Forest Development Plans.

3.2.2 Biological Diversity

Biological diversity issues were addressed through the establishment of stand level reserves as well as landscape level deferrals. In addition to Code reserves, Holberg reported special conservation measures were taken in all 26-engineered blocks and 25 of 26 harvested blocks. Jeune Landing also reported special conservation measures were taken in 2 of 16 blocks harvested (Table 8). These special measures were addressed in the Silviculture Prescriptions.

3.2.3 Soil Stability Maintenance

3.2.3.1 Conservation

Thirty-nine of 60 harvested blocks required special stability assessments and thirty-eight of 64 engineered blocks required special stability assessments. Soil resources were protected at all Operations (Table 9).

3.2.3.2 Operational Site Stabilization Program

Erosion control treatments were carried out at Jeune Landing (3.9 ha) and Holberg (12.5 ha) Operations (Appendix XI). Grass seeding of cut slopes, slides and sensitive sites in current logging blocks is a Company responsibility under the Forest Practices Code. Costs for the site stabilization program totalled $84,500.

Table 9 Protection and Conservation Measures.

Operation	Blocks Engineered	Number of Blocks Requiring Special Consideration								
		Visual Quality	Biological Diversity	Soil Stability	Water Quality	Recreation Resources	Cultural Heritage	Fish Habitat	Wildlife Habitat	Cave/Karst Features
Holberg	26	3	26	16		2	4	26	1	1
Jeune Landing	15	6		10	8			2	2	
Port McNeill	23			12			3	9		
TOTAL	64	9	26	38	8	2	3	37	3	1
Operation	Blocks Harvested	Number of blocks Requiring Special Consideration								
		Visual Quality	Biological Diversity	Soil Stability	Water Quality	Recreation Resources	Cultural Heritage	Fish Habitat	Wildlife Habitat	Cave/Karst Features
Holberg	26	1	25	15			6	26	3	1
Jeune Landing	16	1	2	16	3			4		
Port McNeill	18			8			2	9		
TOTAL	60	2	27	39	3	2	2	39	3	1

3.2.3.3 Watershed Restoration Program

Forest Renewal BC fully funded eight watershed restoration projects in TFL 6 (Table 10). French Creek Forest Services Ltd coordinated the program on behalf of the Company. Accomplishments included 48 km of road deactivation, 2.6 km of fish habitat restoration, and 40 ha of hydro seeding. Program implementation and administration costs were $1,972,589.

Willow wattling is commonly used for stabilizing unstable slopes. Contractor crews from Foresil collected willow material on M1100 road at Port McNeill for use in watershed restoration activities.

Table 10 Watershed Restoration Program Summary

Watershed	Level I	Level II	Road Deactivation	Fish Habitat Restoration	Up-slope Restoration	Spending
Keogh			4.9 km			160576
Hushamu			9.5 km			59619
Goodspeed			18.8 km			564241
Johnston			14.3 km			384571
Marble	✓	✓		1.2 km		225688
Klootchlimmis				0.2 km		40949
Mahatta				1.2 km		42296
Hathaway			0.5 km			23835
NVIR Bio Eng					40 ha	470814
TOTAL	1	1	48.0 km	2.6 km	40.0 ha	$1,972,589

3.2.4 Water Quality

Special engineering to maintain water quality was required for 8 engineered blocks and 3 harvested blocks at Jeune Landing (Table 9).

3.2.5 Recreation Resources

3.2.5.1 Public Recreation Sites

Maintenance of public recreation sites, including provision of firewood and cleaning of facilities was carried out during the summer tourist season. At the Holberg Operation the Hecht Main trail was constructed allowing access to pocket beaches and the coast. A Community Forestry Program crew continued work on the Spruce Bay Trail in the Jeune Landing Operation.

The Company's Public Projects fund covered $35,825 of the recreation site maintenance and upgrading costs. The Forestry Department spent an additional $30,711 on WFP and FRBC recreation projects. Forest Renewal BC funds reimbursed $23,080 of these departmental costs.

3.2.5.2 Recreation Use

There is extensive use of TFL 6 for a variety of recreational activities. Estimates of annual use are in Table 11. These estimates are based on visits recorded to maintained sites, issued permits, and information from the tourism industry. Use of the Holberg sites and trails increases as more people become aware of the recreational areas in and around the Holberg Operation. Koprino was a hot spot for salmon fishing in 1999. The whale watching numbers were down for 1999 as there were very few gray whales seen in the Holberg Inlet during the 1999 season.

Table 11 Recreation Use Estimates

Activity	User Days By Operation			
	Holberg	Jeune Landing	Port McNeill	Total
Beach Use	1700	100	850	2650
WFP Sites and Trails	3600	4000	3500	11100
Hunting	350	800	750	1900
Fishing (Freshwater)	350	2000	1200	3550
Fishing (Saltwater)	1200	2000	1500	4700
Firewood Cutting	350	1000	680	2030
Food Gathering	75	500	250	825
Kayaking	450	300	75	825
Whale Watching	10	0	1000	1010
Auto Touring	9500	4000	1500	15000
Hiking and Caving	750	200	200	1150
TOTAL	18335	14900	11505	44740

3.2.6 Cultural Heritage Resources

All cultural heritage values identified through inventories and assessments are protected during development. When undocumented resources are discovered, operational plans are altered to ensure their protection. Port McNeill Operation modified plans for three future blocks for cultural heritage values in 1999 and Holberg Operation modified four engineered blocks for cultural heritage values (Table 9). Port McNeill modified two harvested blocks and Holberg modified harvest plans in six harvested blocks.

3.2.7 Fish Habitat

3.2.7.1 Conservation

Protection and conservation of fish and fish habitat requires the designation of riparian reserve and riparian management zones at the engineering stage. In 1999, 37 of 64 engineered blocks required special management to protect fisheries values. Thirty-nine of 60 harvested blocks had required similar attention (Table 9). Harvesting requires special falling and yarding techniques near these zones as well as specialized treatments such as aerial pruning to increase windfirmness.

3.2.7.2 Salmonid Enhancement Program

Western Forest Products Limited continues to support Salmonid Enhancement projects in its TFLs. Local volunteers, many of whom are Company personnel, provide volunteer labour and skills to these projects. Hatcheries and habitat improvement projects continue to be a highlight of the forest tours.

TFL 6 watersheds enhanced through the Company's support of salmonid enhancement projects include the Goodspeed and San Josef Rivers near Holberg, the Marble River west of Port McNeill, and the Colonial and Cayegle Creeks south of Port Alice.

In 1999 the hatcheries located within TFL 6 released 1 895 134 fry, and increase of almost 1.5 million over 1998 (Appendix XI). The increase can be credited to an excellent return year resulting in a successful brood stock capture.

The Forestry Department costs associated with the Salmonid Enhancement Program were $100,019. An additional $63,533 came from the Company's Public Projects fund. The federal Department of Fisheries and Oceans reimbursed $15,000 of the Public Projects expenditures.

3.2.7.2.1 Cordy Creek – Holberg

Chinook, coho, chum and steelhead salmon fry were released into the Goodspeed River, and coho salmon fry were released into the San Josef River. A total of 465 510 fry were released from the Cordy Creek hatchery (Appendix XI).

The Holberg forestry department operated the Cordy Creek Hatchery in 1999. The hatchery released more than 465 000 fry, the largest number in the hatchery's history. This success can be attributed to the countless hours donated by Holberg volunteers under the direction of Assistant Forester Jeff Mosher. All fry originated from brood stock captured in the Goodspeed. This was the second consecutive year it was not necessary to supplement with brood stock and eggs from the Marble River Hatchery indicating excellent adult returns to the Goodspeed River. 1999 also proved to be the best year in six years for steelhead broodstock capture and fry release.

3.2.7.2.2 Marble River – Port McNeill

The Friends of the Marble River manage the Marble River Hatchery and Rearing Channel, which are located west of Port McNeill. The 1998 brood stock capture was very high and accounted for the best year for release the Hatchery has seen. The hatchery released 1 045 690 chinook fry, 272 341 coho fry and 3 403 steelhead fry (Appendix XI). The Marble Hatchery also assisted in the raising of 171 752 west coast coho from the Quatse River Hatchery near Port Hardy.

Work on the Rearing Channel continued with the pouring of the walls and the installation of the stop logs and screens. The bottom and the sides of the channel were lined with rock. The Channel was officially opened on June 19, 1999 by the Department of Fisheries and Oceans, Western Forest Products Limited, and the Pacific Salmon Foundation. A ribbon cutting ceremony and the turning of the valve that allows the water flow to the channel marked the opening of the Channel. The Directors and the children present at the ceremony ponded fish in the channel when the water started flowing.

The Friends of the Marble River also worked in the Marble River Watershed with the Watershed Restoration Program In Stream work on the Howlal Creek.

At Marble River volunteers load chinook salmon fry into aluminum containers for helicopter transport to upper reaches of the watershed. The Marble River hatchery is now in its 20th year of operation with dramatic positive impacts on the chinook escapements and adult production.

3.2.7.2.3 Colonial Creek – Jeune Landing

The Colonial Creek Hatchery raised 108 550 coho and chinook that were released into the Colonial watershed in the spring of 1999 (Appendix XI).

After fifteen years of operating the Colonial Hatchery, Dave and Vicky McKinnon of the Port Alice Fish and Wildlife Association are retiring. Western Forest Products appreciates the major contribution they have made to the improvement of the salmon runs in the Colonial watershed. The Jeune Landing Operation - Forestry Department will now be responsible for the operation of the hatchery. The forestry department will still receive help from Dave and Vicky McKinnon and the other volunteers of the Port Alice Fish and Wildlife Association.

3.2.8 Wildlife Habitat

Special measures were taken to address wildlife values in two Jeune Landing harvested cut blocks as well as in one Holberg engineered cut block and three Holberg harvested cut blocks (Table 9). Western Forest Products' management approach is shifting from individual species to a landscape level approach where defined habitat types will be maintained to provide for a range of sensitive species. The Company will continue to inventory and individually manage special wildlife features like raptor nests and bear dens.

3.3 Integration of Harvesting Activities

Western Forest Products encourages the use of the TFL 6 by other stakeholders. Mushroom and salal pickers have access to harvest areas and are only restricted by safety and fire regulations. Yew bark harvest is encouraged in areas under cutting permit. Agreements have been reached with First Nations to focus harvest of traditional materials, mainly cedar bark, in cutting permit areas. Beekeepers use company roads and recreation use is encouraged through the maintenance of existing recreation sites and the expansion of others. However, commercial use of existing public sites is not permitted. Trapline holders are notified when new cutting permits are issued.

3.4 Forest Fire Management

3.4.1 Prevention

Fire Preparedness Plans were submitted to the Ministry of Forests in the spring of the year by all Western Forest Products Operations. Holberg Operation maintained four weather stations and one weather station was maintained by Koprino Logging within the TFL. Two weather stations were maintained by Jeune Landing Operation and Port McNeill Operation maintained three weather stations through the fire season. Total forestry costs for fire prevention and suppression were $14,374.

3.4.2 Suppression

Holberg and Port McNeill reported no accidental fires requiring suppression in the year. Jeune Landing Operation suppressed one accidental fire, which affected less than 0.1 ha of land (Appendix XI).

3.4.3 Fuel Management

The last TFL 6 fuel management plan was produced in 1995 covering the period 1996 through the year 2000. This plan uses fuel type, climatic zone, aspect, and slope to establish a hazard rating that is used to determine fuel management practices. Slash and pile burning are generally not required for hazard abatement purposes. Roadside burning was carried out on 66.4 km of road in 1999 for silvicultural purposes.

3.5 Forest Health Management

Forest health is monitored as part of other regularly scheduled silviculture surveys. Specific forest health costs are only assigned when there is a specific forest health situation such as an insect outbreak.

John Leesing, WFP's former Chief Forester, examines black-headed budworm egg counts on western hemlock needle samples during assessments for the June 1957 aerial spraying of this pest in Quatsino Sound. More than 156,000 acres of northern Vancouver Island were sprayed with a mixture of DDT and diesel at a cost of $252,000 in this B.C./Canada forest protection program

3.5.1 Disease Management

No unusual disease issues were identified during the year. Hemlock dwarf mistletoe and various root rots were noted during silviculture surveys or when preparing silviculture prescriptions and stand management prescriptions. Many of the hybrid poplar clones planted over the last few years in Holberg were killed by poplar leaf bight (*Venturia populina)*. Formal surveys and special management techniques are employed when unusual situations are encountered. None were identified in 1999.

3.5.2 Insect Management

The western black-headed budworm (*Acleris gloverana*) outbreak greatly reduced from the 1998 level in both Holberg and Jeune Landing. Aerial surveys conducted with the MOF showed pockets of light infestation in the Jeune Landing Operation. The overview also showed good recovery in both second growth and old growth damaged by the previous year's infestation. There remained very little development of the black-headed budworm in Coal Harbour or Port McNeill. Costs to monitor the outbreak were $1,040.

Ambrosia beetle (*Gnathotrichus* spp. and *Trypodendron lineatum*) trapping continued at the Port McNeill Dryland Sort. Captures of both species dramatically increased over 1998 levels (Table 12). Higher catches were due to 1998 being an excellent breeding year for the ambrosia beetle. There was a large volume of previously attacked log inventory on-site. Blowdown around the sort bred up a huge population. The costs of the ambrosia beetle-trapping program are covered in logging costs.

Table 12 Ambrosia Beetle Trapping Summary – Port McNeill Dryland Sort

Species	Year		
	1997	1998	1999
Gnathotrichus spp.	2293	750	1750
Trypodendron lineatum	982500	348525	1606605
TOTAL	2748	3749	1610354

3.5.3 Ungulate Management

Deer browsing of regeneration is not a significant problem in most areas of TFL 6. No browse guards or repellents were installed or maintained in the year. Ungulate habitat models were developed and tested during the year. Models were shown to produce acurate results for indicating habitat quality for elk and deer. Ungulate winter ranges remain in place.

3.5.4 Abiotic Factors

Several areas of windthrow were detected during the year. Silviculture Plans were submitted for 6 salvage areas. Several harvested blocks were modified to reduce windthrow. Work began on a study to look at the effects of windthrow on Riparian Areas. Data collection took place in 1999 with analysis to follow in 2000.

3.6 Silviculture

3.6.1 Reforestation

3.6.1.1 Seed Acquisition

Western Forest Products owns seed orchards licensed by the Ministry of Forests to produce genetically superior seed and stecklings. These orchards are located at the Saanich Forestry Centre and the Lost Lake Field Operation on south Vancouver Island. Consolidation of the Lost Lake seed orchards to the Saanich Forestry Centre continued.

Seed orchards that were managed for crop production in 1999 included two western red cedar orchards, one western hemlock orchard, and two coastal Douglas-fir orchards. All other orchards are either too young for seed production, or adequate seed is stored for future use.

The 1999 western red cedar crop was among the largest on record for the Lost Lake orchards, despite significant frost damage. An estimated 14 million seedlings will be grown from the seed produced by 93 hectalitres (hl) of cones harvested (Appendix XXII). A portion of the red cedar crop resulted from supplemental mass pollination or from tops, where self-pollination was expected to be low. Funding for the supplemental mass pollination was received through the Forest Genetics Council's Operational Tree Improvement Program from Forest Renewal BC. Ongoing DNA studies will determine if the out-crossing rates confirm the projected volume gain at age 60. Seed from these collections is appropriate for reforestation in TFL 6 to elevations of up to 745 m.

The coastal Douglas-fir seed extracted from 50 hl of cones is projected to yield 0.8 million seedlings (Appendix XXII). The bulk of the crop resulted from supplemental mass pollination, which raised the projected volume gain at age 60 to 10%. This seed is suitable for use in TFL 6 to elevations of 700 m.

Only the high-value clones from the low-elevation western hemlock orchard were included in the 1999 harvest of 8.5 hl of cones. The projected yield is 1.7 million seedlings that are suitable for reforestation on TFL 6 to elevations of 400 m (Appendix XXII). Harvest from the high-elevation western hemlock orchard produced 6.6 hl of cones for an estimated 1.2 million seedlings. This seed is appropriate for reforestation in TFL 6 between elevations of 436 m to 1036 m.

Cuttings from the yellow cypress clonal hedges at the Lost Lake Field Operation yielded 197.8 thousand stecklings.

No wild stock seed collections were made within TFL 6 during the year. However, wild stock seed collections to fill seed needs within TFL 6 were made. Two western red cedar collections on central Vancouver Island will provide seed for an estimated 1.1 million seedlings appropriate for use in TFL 6 at elevations up to 1 300 m.

Future seed production for western hemlock was seriously set back when vandals severely damaged 71 per cent of the trees in the upcoming high-gain low-elevation orchard at the Saanich Forestry Centre. Attempts to salvage the orchard are underway.

The prorated share of seed acquisition costs for TFL 6 was $196,708 including costs of wild seed collections.

WFP staff using a Girette hydraulic lift to pick cedar cones at Lost Lake. The 1999 western red cedar crop was excellent and replenished the seed supplies with some surplus.Subsequent tests indicated high viability (> 90% germination).

3.6.1.2 Tree Improvement and Orchard Consolidation

The Lost Lake orchards have been declared surplus as of December 1999. While the Company still owns the property, available crops of market value will be managed and harvested. Removal or replication of all stock of value in every species is now complete, with transplanting of numerous yellow cypress hedges to Saanich Forestry Centre, and the collection of scion material from other species.

Replicates of 19 weevil-resistant Sitka spruce received from the Ministry of Forests have been established at the Saanich Forestry Centre replacing non-weevil resistant stock in orchard 172.

Evaluation of low-elevation western red cedar stock continued with breeding in the Lost Lake orchards for the Ministry of Forests breeder. Seed from the breeding work will be grown and outplanted in trials to determine the value of each parent. The Company was awarded funds through the Forest Genetics Council's Operational Tree Improvement Program from Forest

Renewal BC to replicate all western red cedar parents in trials through an extensive grafting project planned for 2000.

Breeding work continued in the high-elevation western hemlock orchard at Lost Lake. The Ministry of Forests' breeder has confirmed that progeny tests will be established to evaluate the gain at rotation delivered by each tested parent. Results from low-elevation parent progeny tests have led to the inclusion of additional selections for the low-elevation western hemlock orchard.

Evaluation of previously established yellow cypress clonal trials continued. High and low elevation replicates of phases 1, 2, and 3 clonal trials are located in TFL 6, together with all replicates of phases 4 to 8, and on replicate of phase 9. From these trials, the top performers will be identified and included in the advanced generation orchards. Rejuvenation of the top clones from phases 1, 2, and 3 selected after analysis of the 4-year data continued with the production of the second of three serial rejuvenations. Top clones from the Ministry of Forests' tests were received and established in hedges for future steckling production.

The share of costs for tree improvement for TFL 6 totaled $54,247. The prorated share of reimbursements from Forests Renewal BC via the Forest Genetics Council's Operational Tree Improvement Program was $53,374.

Saanich Forestry Centre's seed orchards and cypress steckling hedges were attacked by vandals in October. More than 1000 orchard trees were damaged or destroyed as well as 2000 rooted yellow cedar cuttings pulled in a vain attempt to damage what was thought to be genetically modified trees. WFP has never been involved in any GMO work with trees or tree genes.

3.6.1.3 Site Preparation

Site preparation tools on north Vancouver Island include broadcast burning, pile burning, and mechanical site preparation. These techniques are employed to improve plantability or reduce brush competition. Appendix XI details the 1999 site preparation program. Costs for site preparation in TFL 6 were $54,247.

3.6.1.3.1 Broadcast Burning

There was no broadcast burning in TFL in 1999. Although burning was planned it was not possible due to poor weather conditions. Smaller cutblock sizes and the increased number of riparian reserves have made prescribed burning both difficult and expensive. This is the third year in the past thirty where broadcast burning was not used as a silvicultural tool.

3.6.1.3.2 Pile Burning

Roadside debris was piled and burned at all Operations in TFL 6: Port McNeill completed 9.6 km of roadside burning, Holberg completed 50.8 km, and Jeune Landing completed 6.0 km for a total of 66.4 km (Appendix XI). Average cost was $108 per km.

3.6.1.3.3 Mechanical Site Preparation

Holberg and Port McNeill used excavators to prepare planting spots on 25.4 ha of harvested area (Appendix XI). Costs for mechanical site preparation averaged $2,739 per ha.

3.6.1.4 Planting

Silviculture contractors planted 1.08 million seedlings on TFL 6 during the year. Spring planting was delayed in 1999 due to heavy snowfall accumulations that did not melt until late into the year. Original plantations were established on 983.2 ha and fill-plants or replants carried out on an additional 46.0 ha of previously planted area (Appendix XI). The NSR area recorded on the TFL 6 Regeneration Balance sheet increased during the year (Appendix XII). Appendix XIII summarizes the planting history of TFL 6.

Fertilization at time of planting improves the competitiveness of conifer seedlings on salal-dominated sites. Western Forest Products Limited fertilized 415 000 seedlings at time of planting in 1999.

The bulk of the seedlings planted on TFL 6 were western red cedar (42 per cent). Other major species were amabalis fir (33 per cent) and western hemlock (14 per cent). The planting program also included small numbers of cypress, Douglas-fir, Sitka spruce and shore pine.

Costs for the planting program totalled $1,032,022. This averages to $1,023 per ha or $0.96 per tree. This included stock purchase and inspection, planting costs and fertilization at time of planting costs. Forest Renewal BC's Backlog Program reimbursed the Company $5,314 for areas logged prior to October 1, 1987 that remain a government responsibility.

Workers at Saanich Forest Centre nursery package red cedar plug seedlings in bundles of 25 for boxing and shipment to forestry field operations. A total of 1 088 000 seedlings were planted in TFL 6 in 1999.

3.6.1.5 Stocking Surveys

Surveys of naturally regenerated areas at Holberg, Jeune Landing, and Port McNeill showed high success rates. Of 186.9 ha surveyed all were satisfactorily restocked (Appendix XI). The natural regenerated area is credited to the regeneration balance sheet (Appendix XII).

Ninety-seven percent of the plantation area surveyed met or exceeded TFL 6 stocking standards (Appendix XI). Only 44.4 ha of a surveyed 1 757.2 ha was not satisfactorily regenerated. The NSR area is debited to the regeneration balance sheet (Appendix XII) and scheduled for treatment. The Not Satisfactorily Regenerated balance for TFL 6 was 1 939 ha as of December 31, 1999.

Costs for stocking surveys in TFL 6 were $103,423 or $59 per ha. This high cost can be attributed to a major review and re-write of the Stocking Survey Practices and Procedures. The FRBC reimbursements for stocking surveys was $3,780 for areas logged prior to October 1, 1987 that remain a government responsibility.

3.6.1.6 Planting Survival Assessments

Second year survival assessments of planted areas showed high survival rates, exceeding 90 per cent for all Operations. Survival assessments were conducted on a total of 1 447.9 ha of TFL 6 (Appendix XI).

Survival assessments cost $37,565 or $26 per ha. FRBC reimbursements for areas logged before October 31, 1987 were $205.

3.6.1.7 Free Growing Surveys

Both Port McNeill and Holberg Operations carried out free growing surveys during the year. A total of 2 348.3 ha were surveyed. Eighty-three percent of the surveyed area (1 949.3 ha) met Management Plan free growing standards. Only minor treatments are necessary to declare the balance free growing. (Appendix XI).

Costs for free growing surveys were $70,675 or $30 per ha. High survey costs in 1999 are a result of costs associated with finishing final compilation of surveys claimed in 1998. Forest Renewal BC reimbursed the Company $32,069 under the Backlog Forestry Program for areas logged before October 31, 1987.

3.6.2 Stand Management

3.6.2.1 Juvenile Spacing

All Operations implemented spacing programs during the year (Appendix XI). Silviculture contractors spaced a total of 317.5 ha of dense regeneration on TFL 6. Costs for spacing were $661,752 or $2,084 per ha. Forest Renewal BC reimbursed the Company $659,392 for areas spaced under the Enhanced Forestry Program.

3.6.2.2 Brushing and Weeding

Silviculture contractors carried out brushing and weeding treatments on 759.7 ha of forest land. A variety of brushing and weeding techniques successfully reduced alder and salmon berry competition (Appendix XI). Costs of the brushing and weeding program were $543,232. Forest Renewal BC reimbursed the company $384,514 for brushing and weeding for areas logged before October 31, 1987.

3.6.2.3 Pruning

Port McNeill Operation implemented a small pruning program during the year. Silviculture contractors pruned 185.3 ha of previously spaced stands (Appendix XI). Costs for pruning totalled $428,137 or $2 310 per ha. Forest Renewal BC reimbursed the company $424,020 under the Enhanced Forestry Program.

3.6.2.3 Fertilization

All Operations carried out fertilization programs over the year. A total of 3 415.9 ha were fertilized in TFL 6 (Appendix XI). This year's program was the largest in the company's history. Total fertillization expenses were $1,466,451. Forest Renewal BC reimbursed the Company $1,466,258 under the Enhanced Forestry Program.

Northern Mountain Helicopters mobilize for the aerial forest fertilization program in Block 2 at a Misty 100 loading site. This FRBC sponsored program saw 3416 hectares fertilized with 2.5 million kilograms of fertilizer.

3.7 Roads and Bridges

3.7.1 Construction

Road building in 1999 included 87.1 km of new construction and 28.8 km of rebuilt road (Table 12). Seven bridges were constructed, including a 28.5 m steel and concrete structure with a cost of $108,000. This structure replaced a 20 m spruce stringer bridge with a 7 m cedar jump span on the San Josef River within the Holberg Operation. Jeune Landing also built a 21 m bridge over the Johnson creek. Road construction in TFL 6 was halted for the year on Nov 5, 1999 due to shortage of road capital in other regions. Remaining North Vancouver Island Region road capital was moved to other regions.

Table 12 Road and Bridge Construction Summary

Operation	New Roads (km)	Rebuilt Roads (km)	Bridges (#)
Holberg	42.4	3.7	1
Jeune Landing	25.5	19.3	1
Port McNeill	19.2	5.8	5
TOTAL	87.1	28.8	7

3.7.2 Maintenance

Eight hundred and fifteen kilometres of road were maintained in TFL 6. Appendix XI provides a summary by Operation.

Mike Dietsch, Resident Forester, Holberg Forest Operation, with an experimental planting of 3 year lodgepole pine on rehabilitated and recovered roads. Lodgepole pine fills a specific ecological niche on the extremes of the CWH wet subzone.

3.7.3 Current Deactivation

Temporary, semi-permanent, and permanent deactivations of roads are obligations in current areas. Operations deactivated 130.5 km of road during the year: 38.9 km was temporarily deactivated, 52.7 km was semi-permanently deactivated, and 38.9 km was permanently deactivated. Appendix XI lists the amount of deactivated road by level of deactivation and Operation.

3.8 Employment and Economic Opportunities

3.8.1 Direct Employment

The following sections outline direct forest management and manufacturing employment for TFL 6. The manufacturing employment is for the TFL 6 volume processed in Western Pulp or Doman mills and does not include direct employment generated through log sales or trades.

3.8.1.1 Planning, Engineering and Road Development

Planning, engineering and road development for TFL 6 took place at each Forest Operation, in the Port McNeill Regional office and at the Company's head office in Vancouver. Planning and development at the regional level included implementation of the watershed restoration program and local contract mapping for development plans. Planning, engineering and road development employed 104 company and contract employees for 13297 days (Appendix XV – A). Twenty-eight percent was company personnel, with the remainder contractors.

3.8.1.2 Harvesting

Harvesting employment includes the support staff at each Operation. Sixty-eight percent of the 466 people employed in harvesting are local residents and seventy percent are contract employees. TFL 6 harvesting person-days totalled 60 961 (Appendix XV – A) for the year.

3.8.1.3 Transportation

Logs were barged or towed from northern Vancouver Island to processing facilities and markets in the south. Seaspan International held the barging contract and Pacific Towing the towing contract. Log movement generated 6 981 person-days of employment (Appendix XV – A).

3.8.1.4 Processing

Nine Doman and Western Pulp mills processed wood from TFL 6. These were Duke Point Sawmill, Ladysmith Sawmill, Cowichan Bay Sawmill, Silvertree Sawmill, Squamish Sawmill, Nanaimo Sawmill, Tahsis Sawmill, Nanaimo Log Merchandising, and the Port Alice Pulp Mill. Chips from the sawmills were sent to WP's Squamish Pulp Mill. A portion of the TFL 6 volume was either sold or traded to other forest companies or manufacturers.

Prorated direct employment based on wood flow and consumption estimates totalled 129 818 person-days (Appendix XV – A). This total does not include employment generated by TFL logs that were sold or traded to other manufacturers.

3.8.1.5 Silviculture and Integrated Resource Management

Basic and enhanced silviculture, and integrated resource management projects employed 394 people throughout the year at the Operations and Region. Seventy-seven percent were employed by contractors. Nine percent of the basic and enhanced silviculture employees were local residents who generated twenty-eight per cent of the person-days (Appendix XV – A). The TFL 6 prorated share of the Saanich Forestry Centre employment (65 full time or seasonal employees) was 2 337 person-days. Silviculture and Integrated Resource Management generated 13 883 person-days of employment in the year. Western Forest Products' policies continued to focus on the development of First Nations and other local skilled crews.

Saanich Forestry Centre conducts tree improvement as well as *producing nursery stock.*
Kathy Brown, Orchard Technician grafts scion material from improved cedar trees onto cedar root stock for eventual orchard planting The seed orchard program is expanding and being modified to fit new corporate needs following the acquisitions of Crown tenures from Pacific Forest Products.

3.8.1.6 Administration

Administration of the TFL is carried out from the Forest Operation, Regional and corporate offices. Thirty- seven Company employees spent a portion of their time in the administration of TFL 6. The pro-rated share of administrative employment for TFL 6 was 4 261 person-days for the year.

3.8.2 Indirect Employment

TFL 6 activities support secondary jobs in the local communities of Holberg, Port Alice, Port McNeill, Coal Harbour, Quatsino, Winter Harbour, and Port Hardy. While these communities are slowly diversifying their economies, they all still rely heavily on the forest industry.

Central Island, south Island and lower Mainland communities also benefit from the timber produced from TFL 6 through secondary jobs. Significant income is generated in the processing locations: Nanaimo, Ladysmith, Cowichan, and Vancouver. Additional indirect and induced jobs are supported province-wide through services and supplies.

Indirect job opportunities are supported by TFL 6 through the spending of wages from direct jobs. This spending supports jobs in the service, retail, finance, and housing industries.

The Provincial Government stumpage and royalty revenue from this tenure was about $39 million in 1999. Both the Federal and Provincial Governments collected significant revenue from income and other taxes as a result of TFL 6 logging, forest management, and timber processing. Municipal Governments also receive revenue through taxes from TFL 6 operations within their jurisdictions. Indirect jobs within Government Ministries and agencies at all levels are supported by this revenue.

The ratio of direct employment to indirect and induced employment used in this analysis is 1:2 (Price-Waterhouse). For every employee directly supported by TFL 6 activities, two additional people are indirectly employed through private and public sector agencies. The estimated number of person-days of indirect and induced employment generated by TFL 6 activities in 1998 was 458 402. At 180 person-days per year, TFL 6 activities supported more than 2 546 full-time equivalent indirect jobs.

Workers hose down bunkhouses and other equipment on the floating logging camp at Holberg in 1948. The dock and wharf are still located in the same location but the townsite moved on land in the 1960's using houses and other buildings from the old Port Alice townsite which was located adjacent to the pulpmill.

3.8.3 First Nations Employment and Initiatives

The Company continued to work closely with the Quatsino First Nation on their phase logging contract. A total of 44 700 m3 were harvested by the Quatsino Forestry Company. Western Forest Products continued to assist the Quatsino First Nation with the management of Woodlot Number 72 which is encompassed by TFL 6.

The Company continued to work on expanding its relationship with the Kwakiutl First Nation. WFP supported a new position for a Kwakiutl First Nation Forestry Coordinator. The coordinator worked with the Company and band on implementing training initiatives, hiring programs and silviculture programs.

There was a change in the First Nation silviculture contracting from past years. Wallas Forestry chose not to continue with their contracting venture. A combined Quatsino and Kwakiutl contracting company was formed. The new band endorsed company received several direct award contracts in spacing, pruning and brushing.

Silviculture contracts in TFL 6 provided 2 157 person-days of First Nation employment in 1999. This was more than 30 per cent of the contract silviculture work on the tenure. The

Company target of providing a minimum of 20 per cent of the silvicultural contract employment on a tenure to First Nations was exceeded. Appendix XV-C provides First Nations silviculture contract information.

3.9 Performance Monitoring

The Ministry of Forests carried out 260 harvest inspections in TFL 6 during the year. 258 were found to be in compliance and only minor instructions were issued. The two inspections which were not in compliance resulted in a warning ticket being issued in one case and no formal actions in the other case. Four determinations were finalized for infractions that took place in 1997. As an indication of the Company's due diligence the determinations resulted in minimal fines totaling $4,080.

The company changed its internal auditing program during the year. After completing audits on 14 cutblocks and 26.2 km of roads within the Holberg and Jeune Landing Operations the external auditors were replaced with an internal team.

Audits performed by external auditors found that road construction was generally well done with a few minor issues surrounding ditching and culverting. Cutblocks were also generally well managed with improvement in culvert care over previous years. However, some issues around site disturbance, litter and maximizing timber development were identified. Action Plans were developed and implemented to deal with any issues.

The internal audit team conducted audits on Holberg, Port McNeill and Jeune Landing against the implementation of the updated Environmental Management System. Through the audits 8 non-conformance items were identified. Auditors discovered a few workers without maps and a few petroleum spill and handling issues. Appropriate Corrective Preventative Action Reports were developed for each item and action plans have been implemented to address the findings.

4.0 TIMBER PROCESSING

An estimated 1 207 000 m^3 of TFL 6 logs were processed at seven Company owned mills on the lower mainland and southern Vancouver Island as well as at the Port Alice Pulp Mill on Northern Vancouver Island. An additional 169 000 m3 of TFL 6 logs were sold or traded on the Vancouver Log Market. The following mills received logs from TFL 6 in 1999:

- Port Alice Pulp Mill - 263 000 m^3 or 46% of its wood needs for the year.
- Duke Point Sawmill - 460 000 m^3 or most of its wood needs for the year.
- Ladysmith Sawmill - 237 000 m^3 or 67% of its wood needs for the year.
- Silvertree Sawmill – 143 000 m^3 or 49% of its wood needs for the year.
- Tahsis Sawmill – 50 000 m^3 or 15% of its wood needs for the year.
- Cowichan Sawmill – 33 000 m^3 or 13% of its wood needs for the year.
- Nanaimo Log Merchandizer – 16 000 m^3 or 2% of its wood needs for the year.
- Nanaimo Sawmill – 5 000 m^3 or 1% of its wood needs for the year.

The Chemainus Value-Added Mill re-manufactured 114 thousand cubic metres of wood during the year. Some of this volume came from these facilities. The Squamish Pulp Mill consumed 734 000 units of chips. Eleven per cent of this volume came from TFL 6 logs manufactured at these mills.

Estimates of log flow and wood consumption are summarized by tenure in Appendix XVII.

5.0 RESEARCH

Western Forest Products continued to research a variety of forestry issues in 1999. Additionally, the Company supported a number of university projects at the University of British Columbia, Simon Fraser University, and the University of Victoria with funds and in-kind support. Data collections and reports completed in the year are noted in the following sections. All active projects in TFL 6, including those not scheduled for monitoring in the year, are listed in Appendix XVIII.

Forestry Department research costs for TFL 6 totaled $186,564 including the prorated share of the Saanich Forestry Centre's tree improvement research program. Reimbursements totaling $123,126 were received from Forest Renewal BC via the Science Council.

5.1 Silviculture and Stand Management Research

5.1.1 Salal Cedar Hemlock Integrated Research Program (SCHIRP)

Crop tree growth on salal-dominated sites on northern Vancouver Island can be improved through fertilization, as proven by numerous research trials and surveys. The SCHIRP research team includes corporate, federal and provincial government, and University personnel. Western Forest Products has been a leader on this research team for more than a decade. Findings from the SCHIRP trials and surveys have led to the implementation of extensive fertilization programs. Funding for portions of this work was received from Forest Renewal BC through the Science Council.

Numerous survey plots were established in areas receiving operational fertilization treatments in 1999. In areas receiving a second fertilization treatment, measures of previously established plots were recorded. Operational fertilization monitoring plots established in the spring of 1997 were measured before the 1999 flush, thereby including two post-fertilizer growing seasons. Monitoring plots established in the spring of 1995 were measured after the 1999 growing season, thereby including five post-fertilization growing seasons.

Measures of the stock type by fertilization trial on S1CH sites after two growing seasons have been analyzed. For western red cedar, stock type 415s without fertilizer at time of planting outperformed stock type 313s with fertilizer. However, for western hemlock, 415s without fertilizer cannot keep up with the growth of 313s with fertilizer at time of planting.

The intended establishment of a trial on *Vaccinium*-dominated sites at Holberg during 1999 was set back by one year because of an excess of snow on the study site. The layout of the trial has been completed, and planting will take place in 2000. A replicate of the trial was established at Jordan River in the fall of 1999.

Dr. John Barker, who worked for Western Forest Products for more than 30 years, in Block 202 on West 83D in the SCHIRP scarification trial. This hemlock tree was planted and fertilized in the spring of 1996 with 28 grams of fertilizer. John Barker retired from WFP to hold the Chair in Silviculture supported by FRBC at the University of British Columbia. John's many accomplishments with WFP in research and technical forestry services will be long remembered.

5.1.2 Drainage Trial

Maintenance and monitoring of the Suquash Main drainage trial continued in 1999. Previously established vegetation plots were re-visited to record species presence, abundance, and size where applicable. Heights of the western red cedar are showing response to the drainage treatments. Two growing seasons after implementation of drainage, the trees closest to the ditches have the greatest height increments, while trees in subplots further from the ditch have decreasing height increments. Lowest average height increments were recorded for the control plots. At time of drainage, no significant differences in heights were detectable. Funding for this work was received from Forest Renewal BC through the BC Science Council.

5.2 Genetics Research

Three replicates of the ninth and final phase of the yellow cypress clonal evaluation trials were established in 1999, with one of the trials at Port McNeill. Replicates of all nine phases now occur in TFL 6.

Measurements of the third and fourth phases at Jeune Landing and Port McNeill, after seven growing seasons, and of the seventh phase at Holberg, after four growing seasons, were recorded in 1999. Data screening and summarization of the second phase measures after seven growing season were completed. Top performers have been rejuvenated in the tree improvement program. Funding for this work was received from Forest Renewal BC through the Science Council.

Don Pigott, a WFP contractor and owner of Yellow Point Propagation Ltd., works on Centre Main near Port McNeill to apply gibberellic acid for cone induction on natural cypress regeneration. This trial is to assess feasibility of increased natural cone production in this important species. Most planting stock in Cy is from stecklings because of poor natural seed production.

5.3 Forest Inventory

Work on Vegetation Resource Inventory continued in 1999. Photo interpretation and quick plot ground sampling were completed. A more intensive ground survey plan is planned for 2000.

6.0 GOALS AND INITIATIVES

Western Forest Products manages TFL 6 consistent with the objectives outlined in Management Plan 8. Appendix XX outlines progress made in addressing Management Plan 8 commitments and issues in 1999. In 2000 draft Management Plan 9 will be submitted to MOF, public, and stakeholders for review and comment. For 2000 there are specific silviculture, planning, inventory, research, recreation, First Nations and market goals.

The silviculture goals are to complete all basic silviculture obligations according to operational plan schedules as well as to pursue opportunities to expand the current enhanced silviculture programs. Program details are provided in Table 14.

Table 14 Goals for 2000

Silviculture	Site Preparation Planting Survival Surveys Stocking Surveys Brushing Free Growing Spacing Pruning Fertilization	100 ha 1200 ha 1700 ha 2440 ha 942 ha 5200 ha 177 ha 128 ha 60 ha
Planning	De-centralize mapping services Landscape Unit Planning Submit MP 9	TFL 6 Blocks 1 and 2 TFL 6 Blocks 1 and 2 TFL 6 Blocks 1 and 2
Inventory	Wildlife Modelling Fisheries Inventory VRI Phase 2	Special Management Zones Block 2 Watershed TFL 6 Blocks 1 and 2
Research	SCHIRP Monitoring Suquash Monitoring CRIC (phase 9) Installation Windthrow Research	TFL 6 Block 2 TFL 6 Block 2 TFL 6 Block 2 TFL 6 Blocks 1 and 2
Recreation	SMZ # 2 Development Recreation Site Maintenance	TFL 6 Block 1 TFL 6 Block 1 and 2
First Nations	Capacity Building	TFL 6 Blocks 1 and 2
Markets	Certification – FSC and ISO	TFL 6 Blocks 1 and 2

In 2000 plans are to move FDP mapping functions from a local contractor to a company employee based in Port McNeill. We plan to submit draft Management Plan 9.

In 2000 Forest Inventory efforts will continue with Phase 2 of the re-inventory of TFL 6 forests. Wildlife surveys will continue with detection work on marbled murrelets.

Current research projects will be maintained. There will be continued monitoring of SCHIRP sites; installation of mid-elevation SCHIRP style installations to examine Vaccinium competition, continue monitoring of the Suquash drainage site and further research into windthrow management and estimates of windfall losses.

Recreation goals will be to maintain existing WFP sites within TFL 6 as well as to continue cautious development of SMZ # 2 while keeping in mind the SMZ recreation objectives. There will be an attempt to resolve Raft Cove public access issues.

First Nations efforts will be concentrated on continued capacity building with both the Quatsino and Kwakiutl First Nations. The Quatsino First Nation currently logs, does silviculture work and some assessments for WFP. The Kwakiutl First Nation currently does silviculture work and has some members training and employed in logging for WFP.

Certification remained a high priority during 1999. A significant amount of time was dedicated to ensuring the company's readiness for FSC and ISO field assessments. Field assessments are expected to take place in early 2000. The Company will continue its efforts to obtain FSC, ISO and CSA certification for TFL 6 woodland operations in order to aid marketing of forest products in foreign markets.

7.0 ADMINISTRATION

The costs of supervision and overhead relating to specific operational projects are included in the reported project costs. However, many administrative costs are broadly defined as general management and overhead. These include Company prorated charges to the Forestry Department for rentals and services as well as salaries and expenses to perform administrative functions. Total costs for administration were $709,301.

8.0 FINANCIAL STATEMENTS

8.1 Forest Management Costs

The following summary presents all TFL 6 Forest Management Costs incurred or expended by Western Forest Products and other agencies at the field level. Planning, supervision, and overhead costs directly related to programs as well as program application costs are included before credits or reimbursements. A prorated share of head office forestry department and Saanich Forestry Centre costs are included as well.

Program	Section		Sub-Total
Planning			$611,317
	Higher Level Planning	129,892	
	Forest Development Planning	87,809	
	Cutting Permits and CP Cruising	263,572	
	Silviculture Prescriptions	130,044	
Public Involvement			$75,975
	Forest Education	75,975	
Inventories and Mapping			$515,901
	Geographic Information System	105,642	
	Forest Inventory	170,444	
	Ecosystems	6,389	
	Terrain Stability	3,491	
	Integrated Resource Management	229,935	
Utilization			$75,124
	Residue Assessments	56,957	
	Minor Products	17,486	
	Commercial Thinning	681	
Conservation and Protection			$274,385
	Operational Site Stabilization	84,500	
	Recreation Resources	67,228	
	Salmon Enhancement Program	100,019	
	Fire Management	14,374	
	Forest Health	1,040	
	Audits	7,224	
Silviculture			$4,664,849
	Seed Procurement	196,708	
	Tree Improvement and Orchard Consolidation	54,247	
	Site Preparation	69,580	
	Planting	1,032,022	
	Stocking Surveys	103,423	
	Survival Assessments	37,565	
	Free Growing Surveys	70,675	
	Juvenile Spacing	661,752	
	Brushing & Weeding	543,232	
	Pruning	428,137	
	Fertilization	1,466,451	
	Species Conversion	1,057	
Research			$186,564
Administration			$709,301
TOTAL			**$38,201,171**

8.2 Forest Management Reimbursements

Western Forest Products Limited received funding from government agencies for silvicultural enhancement, integrated resource management, research and tree improvement, and salmonid enhancement projects. A summary of the reimbursements received is presented below.

Program			TFL 6	TFL 19	TFL 25
Forest Renewal BC	MYA* Backlog Forestry	Surveys	36,054	1,360	3,826
		Planting	5,314	7,777	
		Brushing	384,514		52,914
		Species Conversion			3,624
		Site Stability		14,820	
	MYA Enhanced Forestry	Juvenile Spacing	659,392	701,182	273,278
		Pruning	424,020	229,680	292,534
		Fertilization	1,466,258	213,410	601,519
		Research			1,915
	MYA Operational Inventory	Ecosystems		107,910	31,479
		Cultural Heritage			31,347
		Streams	111,245	15,071	9,474
		Timber	87,182	7,549	188,980
		Wildlife	94,927	95,333	328,034
	MYA Public Relations	Recreation	23,080	149,233	
		Public Relations	1,929	704	5,162
	MYA Administration	Administration	147,284	73,670	83,149
	Forest Genetics Council	Operational Tree Improvement Program	53,374	38,797	14,654
	Science Council	Research	123,126	41,818	26,594
Ministry Contracts		Recreation		62,902	
Department of Fisheries and Oceans		Salmonid Enhancement Program	15,000		
South Moresby Forest Replacement Fund		Juvenile Spacing			149,885
		Integrated Resources Management Research			40,000
TOTAL			15,000	62,902	189,885

*MYA – Multi-Year Agreement

APPENDICES

Appendix I

TREE FARM LICENCE 6
1999 Scaled Production

CUBIC METRES

Operation	Mark	Volume	Total
Botel Lake	6/301	311	
	6/302	54194	
	6/500	3305	
	6/501	9463	
	6/502	11117	
	6/503	826	
	6/504	2554	
	6/505	1083	
			82853
Coal Harbour	6/601	11919	
	6/602	8120	
	KZ003	58	
	T0457A	0	
	T0475A	5047	
	T0475B	22795	
	T0693B	4099	
	6/603	18983	
	T0457B	3349	
	T0457C	6354	
	T0475C	458	
	T0700B	4309	
			85491
Holberg			
	6/102	140	
	6/104	7131	
	6/750	49120	
	6/801	4721	
	6/802	638	
	6/803	60636	
	6/804	13962	
	6/872	6581	
	6/873	48201	
	6/90	5064	
	CZ010	35612	
	T0386A	12907	
	T0386B	13126	
	T0391A	6671	
	T0660A	520	
	6/1	99	
	6/103	11560	
	6/105	35179	
	6/751	57802	
	6/752	32987	
	6/805	11636	
	6/874	39629	
	T0391B	3227	
	T0399B	615	
	T0408A	5963	
	T0445A	5218	
	T0470B	2045	
	T0660B	33277	
			504267

Operation	Mark	Volume	Total
Jeune Landing	6/200	3630	
	6/202	6831	
	6/203	25316	
	6/205	19432	
	6/97	2219	
	T0508A	4114	
	T0514B	18742	
	T0514C	25856	
	T0514D	13357	
	T0526B	17546	
	T0526C	5187	
	6/2	40	
	6/204	6481	
	6/208	6447	
	6/210	474	
	6/605	1624	
			157296
Mahatta River	6/400	8843	
	6/402	55529	
	T0599	2694	
	6/403	24434	
	6/404	50684	
	6/405	5220	
	6/406	3407	
	T0699A	46754	
			197565
Port McNeill	25/4	2419	
	25/402	1920	
	25/403	16	
	25/404	1802	
	25/98	44	
	6/98	3761	
	BZ002	98552	
	HZ004	6794	
	T0191A	517	
	KZ014	1466	
			117291
Rupert Inlet	6/900	44827	
	6/901	36723	
	6/902	39178	
	T0481A	17667	138395
Winter Harbour	6/7	107	
	6/700	167	
	6/701	3575	
	6/702	164	
	6/703	26686	
	T0664A	1254	
	T0664B	24891	
	6/704	34037	
	T0664C	1463	92344
Total Company Tenures			516296
Total Crown			859205
Grand Total			**1375501**
Total Company and Phase Contractor Operations			1121073
Total Harvested under Full Contracts			254428

Appendix II

TREE FARM LICENCE 6
1999 Volume Charged to Allowable Cut

CUBIC METRES

Mark	Crown Grant	Licences	Crown	TOTAL
25/4			2419	
25/402			1920	
25/403			16	
25/404			1802	
25/98			44	
6/102			140	
6/104			7131	
6/200			3630	
6/202			6831	
6/203			25316	
6/205			19432	
6/301			311	
6/302			54194	
6/400			8843	
6/402			55529	
6/601			11919	
6/602			8120	
6/7			108	
6/700			168	
6/701			3576	
6/702			164	
6/703			26686	
6/750			49120	
6/801			4721	
6/802			638	
6/803			60636	
6/804			13962	
6/872			6581	
6/873			48201	
9/90			5064	
6/900			44827	
6/901			36723	
6/902			39178	
6/97			2219	
6/98			3761	
BZ002	98552			
CZ010	35612			
HZ004		6794		
KZ003		58		
T0191A		517		
T0386A		12907		
T0386B		13126		
T0391A		6671		
T0457A				
T0475A		5047		
T0475B		22795		
T0481A		17667		
T0508A		4114		
T0514B		18742		
T0514C		25856		
T0514D		13357		
T0526B		17546		
T0526C		5187		

Mark	Crown Grant	Licences	Crown	TOTAL
T0599		2694		
T0660A		520		
T0664A		1254		
T0664B		24891		
T0693B		4099		
KZ014		1466		
6/1			99	
6/103			11560	
6/105			35179	
6/2			40	
6/204		6481		
6/208			6447	
6/210			473	
6/403			24434	
6/404			50684	
6/405		5220		
6/406			3407	
6/500			3305	
6/501			9463	
6/502			11117	
6/503		826		
6/504			2554	
6/505			1083	
6/603			18983	
6/605			1623	
6/704			34037	
6/751			57801	
6/752			32986	
6/805		11636		
6/874		39629		
T0391B		3227		
T0399B		615		
T0408A		5963		
T0445A		5218		
T0457B		3349		
T0457C		6354		
T0470B		2045		
T0475C		458		
T0660B		33278		
T0664C		1463		
T0699A		46753		
T0700B		4309		
Grand Total	**134164**	**382132**	**859205**	**1375501**
Residue				
Recognized residue survey volumes associated with 1999 MOF S&R invoices				28952
Total Chargeable				**1404453**

Appendix III

TREE FARM LICENCE 6
Area Denuded – 1999

HECTARES

Operation	Crown Grant	Crown / Licence	TOTAL	Medium, Low, and Very Low Ecosites
Holberg	30.2	345.8	376.0	77.4
Koprino		80.5	80.5	
Michelson		38.3	38.3	
Winter Harbour		116.1	116.1	
Botel Lake		58.6	58.3	
TOTAL HOLBERG OPERATION	**30.2**	**639.3**	**669.5**	**77.4**
Jeune Landing		190.8	190.8	38.1
Mahatta		253.3	253.3	4.6
TOTAL JEUNE LANDING OPERATION		**444.1**	**444.1**	**42.7**
Coal Harbour		109.4	109.4	
Rupert Inlet		145.3	145.3	
Port McNeill	108.4	58.6	167.0	69.6
Jeune Landing		1.0	1.0	
TOTAL PORT McNEILL OPERATION	**108.4**	**314.3**	**422.7**	**69.6**
TOTAL	**138.6**	**1397.7**	**1536.3**	**189.7**
SBFEP @ Port McNeill		36.4	36.4	n/a
SBFEP @ Jeune Landing		34.1	34.1	n/a
GRAND TOTAL	**138.6**	**1468.2**	**1606.8**	**189.7**

Appendix IV

TREE FARM LICENCE 6 Current Cut Control Period Annual Allowable Cut		
CUBIC METRES		
Year	**Allowable Cut Available to Licencee**	**Chargeable Cut**
1995	1 198 361	1 326 233
1996	1 275 506	1 325 989
1997	1 275 506	1 267 622
1998 [2]	1 323 084	909 008
1999	1 464 264	1 404 453
TOTAL	**6 536 720**	**6 233 305** [1]

1 The cumulative cut at the end of the fifth year of the cut control period is 95% of the five year AAC of 6 536 720 m^3

2 TFL was amended to include TFL 25 Block 4 (Port McNeill) effective October 1, 1998.

Appendix V

TREE FARM LICENCE 6
Historical Cut Control Performance
1951 – 1999

CUBIC METRES

Cut Control Period	Allowable Cut Available to Licencee	Chargeable Cut
1951 – 1960	4 168 156	4 242 856
1961 – 1965	3 652 867	3 937 601
1966 – 1970	5 703 003	5 749 372
1971 – 1972	2 619 304	1 862 800
1973 – 1977	6 250 735	5 915 767
1978 – 1979	2 361 622	2 544 317
1980 – 1984	6 321 622	6 618 305
1985 – 1989	6 462 600	6 944 146
1990 – 1994	5 906 900	5 941 356
1995 - 1999	6 536 720	6 233 305
TOTAL	10 552	49 999 825

Appendix VI

TREE FARM LICENCE 6
Coastal Contractor Clause Performance Report

CALENDAR YEAR 1999

Reference	Description		Source
1)	Total AAC of TFL approved by Chief Forester (CF) that is available to Licencee	1 464 264 m^3	CF's approval letter for management and working plan
2)	AAC attributable to Schedule "B" lands that is available to Licencee	1 131 994 m^3	Derived from the approved MWP
3)	Volume of timber harvested	1 375 502 m^3	Obtained from the Regional Timber Officer of District Manager; the total volume of timber that is billed to the Licencee under the Licence during the calendar year (Section 49.1 of Forest Act)
4)	Harvested volume attributed to Schedule "B" lands	1 063 374 m^3	Calculated: (#2/#1) X #3
5)	Total volume contracted under full and phase contracts	643 743 m^3	Licencee Records
6)	Total volume contracted expressed as a per cent of compliance required	121.1 %	Calculated: (#5/(#4 X 0.5)) X 100

Licencee Name: Western Forest Products Limited

Completed by: William Dumont, R.P.F.

Date Report Completed: July 27, 2000

Appendix VII

TREE FARM LICENCE 6
Phase and Full Contractors – 1999

Contractor	Phase
AH Jackson Corporation	Full, Poling
C&E Road Building Ltd	Road
CSK Hauling Ltd	Hauling
GCB Ventures Ltd	Hauling
Goodspeed Road Works Ltd	Road
Helgesen Logging Ltd	Hauling
Holberg Contracting Ltd	Road
JF Trucking Ltd	Road
Juthans Trucking Ltd	Road
Koprino Contracting Limited	Road
Koprino Logging Contractors Limited	Full
	Road
Leeson Lake Contracting Ltd	Falling
Lefty's Log Hauling	Hauling
MacLeod & Smith Log	Hauling
Mi Don Contracting Ltd	Hauling
Mike Jones Ent. Ltd.	Road
Poor Boy Enterprises Limited	Road
Port McNeill Enterprises Ltd	Road
Pt. Alice Salvage Ltd	Hauling
Quatsino Forest Company	Loading
	Yarding
Raft Cove Cont. Ltd	Road
Ransom Creek Falling Ltd	Falling
Stan Zwicker Contracting Ltd	Road
Steele Rock Hauling Ltd	Hauling
WD Moore Logging Co. Ltd	Loading
	Road
	Yarding
GLM Falling Limited	Falling
R & K Tyson Contracting Ltd	Hauling
DDH Contracting Limited	Road
531088 BC Ltd	Road
North Island RockPro Inc	Road
557025 BC Ltd	Full

Appendix VIII

TREE FARM LICENCE 6
Contractors – 1999

Name	Work	Location
4 Way Cedar	MP Salvage	Port McNeill
4469773 B.C. Ltd.	Bunk rebuilds (welding)	QDLS
528248 B.C. Ltd. dba Cutting Edge Forestry	Spacing	Jeune Landing
531088 BC Ltd.	Road	Port McNeill
557025 B.C. Ltd.	Stump-to-dump	Mahatta River
A. H. Jackson Corp.	Poling	Port McNeill
Arbor Tech	Terrain Assessments	Port McNeill
Ashaw Contracting	Yarding & Loading	Holberg
Associated Engineering	Engineering Consultants	Port McNeill, Holberg
Azmeth Forest Consultants	Residue Surveyor	Port McNeill
B. Gurney	Log Scaling	Port McNeill
BC Conservation Foundation	Watershed Restoration	Port McNeill
Bellows, Anna	Recreation Site Maintenance	Holberg
Bivouac West Contracting Ltd	Pile Burning (Forestry)	Holberg
Blue Canyon Enterprises	Yew Bark Harvest	Port McNeill
Builders Warehouse Ltd.	Rigging Deliveries	QDLS
C & E Road Building Ltd.	Road Construction	Jeune Landing
C. Cyr	Log Scaling	Port McNeill
Chrystal Menzies	Hatchery	Holberg
Clarke, Danny	Fire Pump Mtnc	Holberg
Coast Dryland Services Ltd.	Log Rehauling	Port McNeill
Coast Range Contracting	Tree Planting	Holberg
Community Forestry Program	Enhanced Forestry	Port McNeill
CSK Hauling Ltd	Log Hauling	Holberg
D. Anderson	Tours	Port McNeill
D. E. Whidden & Assoc Ltd	Engineering Consultants	Holberg
D.D. & H. Contracting Ltd	Road Construction	Holberg
D.H. Timber Towing & Salvage Ltd.	Boom Towing/Barge tending	QDLS
D.K. Trucking & Contracting	Road Construction & Deactivation	Jeune Landing
Dabber Contracting Ltd.	Backhoe services	QDLS
Donna Larade	Janitorial Services	QDLS
Eagle Cedar	MP Salvage	Port McNeill
Fishfor Contracting Ltd	Stream Assessments	Port McNeill, Jeune Landing, Holberg
Foresil Enterprises Ltd	Silviculture Contractor	Holberg
Foresite Consultants	Software Programs	Port McNeill
Fortis Forestry Mgmt Services	Engineering Consultants	Holberg
Free Growing Enterprises Ltd.	Ground Foliar & Roadside Spraying	Jeune Landing
French Creek Forest Services	Watershed Restoration	Port McNeill, Jeune Landing, Holberg
Friesen, Ray	Recreation Site Maintenance	Holberg
G. Wickstrom	Tours	Port McNeill
G.L.M. Falling Ltd.	Falling	Port McNeill, Jeune Landing, Holberg
GCB Ventures Ltd	Log Hauling	Holberg
GDS Contracting Ltd	Yarding & Loading	Holberg

Name	Work	Location
General Tree Planting Ltd.	Stem Injection & Mechanical Alder Control	Jeune Landing
Goodspeed Road Works Ltd	Road Construction	Holberg
H. Forster	Machine Cleaning	Port McNeill
Haugrud, Morris	Steam Cleaner	Holberg
Hayes Forest Services Ltd.	Helicopter Logging	Jeune Landing
Helgesen Logging Ltd	Log Hauling	Holberg
Herb Saunders Contracting	Road Maintenance, Gravel trucks (pit)	Holberg, QDLS
Hobal, Robert	Floor Installation	Holberg
Holberg Contracting Ltd	Road Construction	Holberg
Holberg Salvage & Disposal	Garbage Disposal	Holberg
Independent Flaggers	Flagging	QDLS
Interior Reforestation Ltd.	Hydro & Dry Seeding	Jeune Landing
Iota Construction	Bridge Contractor	Port McNeill, Jeune Landing, Holberg
J & T Silviculture	Enhanced Forestry	Port McNeill
J. F. Trucking	Road Construction	Port McNeill, Jeune Landing, Holberg
J. Forsberg	Grounds Maintenance	Port McNeill
J.R.P. Consulting Ltd.	Forestry Consulting	Jeune Landing
JRP Consulting	Development Plan/ Surveys	Port McNeill
Juthans Trucking Ltd	Road Construction	Holberg
K. Pazarena	Machine Cleaning	Port McNeill
K. Prachnau	Janitor	Port McNeill
Ker Point Log Salvage Ltd.	Deadhead Salvage	QDLS
Koprino Contracting Ltd	Road Construction	Holberg
Koprino Logging Contracting Ltd	Road Construction	Holberg
Koprino Logging Contractors Ltd.	Log Hauling	Jeune Landing
Kwakiutl Training Crew	Enhanced Forestry	Port McNeill
L. Kueber	Log Scaling	Port McNeill
Leanne Boas	Janitorial Services	Jeune Landing
Ledge Point Contracting	Dryland Sort	Port McNeill
Leeson Lake Logging Ltd	Falling & Bucking	Winter Harbour
Lefty's Log Hauling Ltd.	Log Hauling	Jeune Landing
Lemare Lake Logging Ltd.	Log Hauling, Dumping & Booming	Port McNeill, Jeune Landing, Mahatta River
Long Line Logging Ltd	Yarding Labour	Holberg
M.Jones Enterprises Ltd.	Excavating & capping(pit)	QDLS
Mac Load & Smith Log Hauling	Log Hauling	Jeune Landing
Mahatta Cedar Products Ltd.	MP Salvage, General Labor	Port McNeill, Jeune Landing
Mahatta River Logging Ltd.	Skidder Services/MR Dump Contractor	Jeune Landing
Mark Wildeman	1st Aid instruction	QDLS
Marlinspike Enterprises Inc	Yarding & Loading	Holberg
McLaughlin, Evelyn	Janitorial	Holberg
Midon Trucking Ltd	Log Hauling	Holberg
Mike Jones Enterprises Ltd.	Backhoe Work	Jeune Landing
N&R Forest Management Ltd.	Forestry Consultant	Jeune Landing
N. Boniface	Parks Maintenance	Port McNeill
Noel Roddick Ltd	Silviculture	Holberg
North Island Diving & Watersports Ltd.	Carriage Repairs	QDLS
North Island Forestry Service	Enhanced Forestry	Port McNeill
North Island Powerwash Ltd.	Steam cleaning	QDLS
North Island Rock Pro	Road Construction, Blasting	Port McNeill, QDLS
North West Hydraulic Consultants	Engineering Consultants	Port McNeill

Name	Work	Location
Northern Mountain Helicopters	Fertilization, Silviculture	Port McNeill, Holberg
Northern Prime Cuts	MP Salvage	Port McNeill
Northwest Hydraulic Consultants Ltd.	Instream Consultants	Jeune Landing
O. K. Paving	DLS Re-surfacing/Surfacing	Port McNeill, QDLS
Otter Enterprises	Equipment/Misc. repairs	QDLS
P. Burns	Boom	Port McNeill
Pacific Forest Maintenance	Silviculture	Jeune Landing, Holberg
Pacific Rainforest Scaling	Log Scaling	Port McNeill, QDLS
Piteau & Associates Ltd	Engineering Consultants	Port McNeill, Jeune Landing, Holberg
Poor Boy Enterprises Limited	Road	Port McNeill
Port Alice Dredging Ltd.	Dredging	QDLS
Port Alice Logging Co.Ltd.	Loading, Rebundling WFP Boomsticks	Jeune Landing, QDLS
Port Alice Salvage Ltd.	Log Hauling	Jeune Landing
Port McNeill Consulting	Engineering Consultants	Port McNeill
Port McNeill Enterprises Ltd.	Road	Port McNeill
Port McNeill Shake & Shingle	MP Salvage	Port McNeill
Quatsino Cablevision Ltd	Electrician	Holberg
Quatsino Forest Company	Yarding	Port McNeill
R & K Tyson Contracting Ltd	Log Hauling	Holberg
R. Springer	Machine Cleaning	Port McNeill
R.J. Hay Consulting Ltd.	Engineering Consultant	Jeune Landing
Race Enterprises(aka Harper)	General Labour	Jeune Landing
Raft Cove Contracting Ltd	Road Construction	Holberg
Ransom Creek Falling Ltd.	Falling & Bucking	Jeune Landing
Rare Image	Research	Port McNeill
Rene Laviolette	Mechanic	Holberg
Roosevelt Cedar	MP Salvage	Port McNeill
S. VonSchilling	Log Truck Driver	Port McNeill
S. Wainwright	Grounds Maintenance	Port McNeill
Schaefer Contracting Ltd	Yarding & Loading	Holberg
Shawn Hamilton	Bio Engineering Consultant	Port McNeill, Jeune Landing
Simons Reid Collins	Timber Inventory	Port McNeill
Sources Archeological & Heritage Consultants	Archeological Surveys	Port McNeill
Stan Zwicker Contracting Ltd.	Road	Port McNeill
Steele Rock Hauling Ltd	Log Hauling	Holberg
Terra Engineering	Engineering Consultants	Holberg
Timberline Reforestation Ltd	Tree Planting	Port McNeill, Jeune Landing
Tree of Life Essential Oil	Cedar Oil Production	Port McNeill
Vancouver Island Helicopters	Silviculture	Port McNeill
W.D. Moore Logging Ltd	Y & L and Road Construction	Winter Harbour
Wallas Forestry	Enhanced Forestry	Port McNeill
Wardell Logger's Troubleshooting Service Ltd.	Dump rebuild	Mahatta River
Warner Bay Investments Ltd	Yarding & Loading	Holberg
West Coast Helicopters	Silviculture	Port McNeill, Holberg
Western Aerial Applications	Silviculture	Jeune Landing, Holberg
Westrek Geotechnical Services	Eng Consultant	Holberg
Willdon Construction Ltd	Building Contractor	Holberg
Willson, Rebecca	Hatchery	Holberg
Woodstock Management Inc.	Pest Suppression	Port McNeill

Appendix IX

TREE FARM LICENCE 6
Partitioned Harvest Performance Summary – 1999

MEDIUM, LOW, AND VERY LOW SITES[1]

Operation	Cutting Permit Numbers	Logged			
		Productive	Non-Productive	M/L/VL Ecosites[2]	
		Area (ha)	Area (ha)	Area (ha)	Volume (m^3)
Holberg				77.4	35 165
Jeune Landing				38.1	25 441
Mahatta				4.6	3 254
Port McNeill				69.6	31 068
TOTAL				189.7	94 928

1 Performance must be 52 000 m3 minimum per year on a five-year average.
2 Inventory/MP basis only. Volume estimates per hectare are higher than actual production.

Helicopter

In 1999 approximately 8.2 ha were helicopter logged from within the conventional portion of the operability in the Jeune Landing Operation. Total volume harvested was 6 221 m^3.

Appendix X

TREE FARM LICENCE 6
Small Business Forest Enterprise Program
Harvesting Report – 1999

Year	Volume Available to SBFEP m^3	Licence No.	Licensee	Area Logged ha	Volume Scaled m^3	Residue m^3	Chargeable Volume m^3	TS Harvest Status
1988	25800			0	0	0	0	
1989	61673	TS24141	Koskimo Cont.	3.4	0	0	0	Completed 1990
1990	100620	TS24141	Koskimo Cont.	0	2733	256	2989	Completed 1990
		TS35673	Primex	27.0	9649.0	0	9649	Completed 1995
1991	100620	TS35673	Primex	65.4	62813.0	0	62813.0	Completed 1995
1992	100620	TS35673	Primex	101.3	84958.0	0	84958.0	Completed 1995
		TS37073	Hudson Mitchell	0	0	0	0	Completed 1997
1993	100620	TS35673	Primex	114.1	72571.0	6952.0	79523.0	Completed 1995
		TS37073	Hudson Mitchell	54.9	74016.0	1127.0	75143.0	Completed 1997
		TS39536	BW Creative Wood	1	1003	0	1003	Completed 1993
1994	100620	TS35673	Primex	65.3	62274.0	2209.0	64483.0	Completed 1995
		TS37073	Hudson Mitchell	48.0	23959.0	0	23959.0	Completed 1997
1995	100620	TS35673	Primex	0	594.0	0	594.0	Completed 1995
		TS37073	Hudson Mitchell	88.0	51075.0	1396.0	52471.0	Completed 1997
1996	0	TS37073	Hudson Mitchell	91	60985.0	985.0	61970.0	Completed 1997
1997	0	TS37073	Hudson Mitchell	0	4412.0	5328.0	9740.0	Completed 1997
1998	19373	TSL 43355	BW Creative Wood	25.3	22526.0	0	22526.0	In Progress
1999	19373	TSL 43354	Lukwa Mills	34.1	29314	0	29314	In Progress
		TSL 43355	BW Creative Wood	36.4	30746	0	30746	In Progress
TOTAL	**729939**			**755.2**	**593628**	**18253**	**611881**	

Appendix XI

Project Summary – 1999						
Project			Port McNeill	Holberg	Jeune Landing	TOTAL
Denudation	Total Company	ha	422.7	669.5	444.1	1536.3
	Crown / Licence	ha	314.3	639.3	444.1	1397.7
	Crown Grant (MF)	ha	108.4	30.2		138.6
	SBFEP	ha	36.4	0	34.1	70.5
	TOTAL	ha	459.1	669.5	478.2	1606.8
Accidental Fires		No.			1	1
		ha			< 0.1	0.1
Site Preparation	Prescribed Burning	ha				
	Pile Burning	km	9.6	50.8	6	66.4
	Mechanical	ha	15.5	9.9		25.4
	Crown / Licence	ha		9.9		9.9
	Crown Grant (MF)	ha	15.5			15.5
Planting	Original	ha	295.2	464.8	223.2	983.2
	Replants	ha	0.5			0.5
	Fill-ins	ha	1.0	22.7	21.8	45.5
	Total	ha	296.7	22.7	245.0	564.4
	Crown / Licence	ha	201.7	487.5	245.0	934.2
	Crown Grant (MF)	ha	95.0			95.0
	Number of Seedlings	Cw	176885	252977	22920	452782
		Hw	58900	78161	11350	148411
		Ba	48111	135591	177849	361551
		Fdc	21280	1240	2610	25130
		Yc	29931	13880	6230	50041
		Ss	6180	29538	13003	48721
		P		1400		1400
		Misc				
		Total	341287	512787	233962	1088036
Stocking Surveys	Natural Regeneration	ha	3.1	0	186.9	190.0
	Sufficiently restocked	ha	3.1	0	179.8	182.9
	Crown / Licence	ha	3.1	0	179.8	182.9
	Crown Grant (MF)	ha				
	Other Classification	ha			7.1	7.1
	Crown / Licence	ha			7.1	7.1
	Crown Grant (MF)	ha				
	Plantation Regeneration	ha	528.8	745.8	482.6	1757.2
	Sufficiently Restocked	ha	498.5	742.5	471.8	1712.8
	Crown / Licence	ha	247.1	742.5	471.8	1461.4
	Crown Grant (MF)	ha	251.4			251.4
	Not Sufficiently Restocked	ha	30.3	3.3	10.8	44.4
	Crown / Licence	ha	4.6	3.3	10.8	18.7
	Crown Grant (MF)	ha	25.7			25.7
	Other Classification	ha				
	Crown / Licence	ha				
	Crown Grant (MF)	ha				
Silviculture Prescription		ha	224.9	579	665	1468.9
		No.	11	16	20	47
Plantation Survival Assessments		ha		837.8	610.1	1447.9
		%		90.8		90.8

Project			Port McNeill	Holberg	Jeune Landing	TOTAL
Free Growing Surveys	Total	ha	1936.8	411.5		2348.3
	Free Growing	ha	1542.8	406.5		1949.3
	Crown / Licence	ha	1297.9	406.3		1704.2
	Crown Grant (MF)	ha	244.9			244.9
	Not Free Growing	ha	394.0	5.2		399.2
	Crown / Licence	ha	394.0	5.2		399.2
	Crown Grant (MF)	ha				
	Other Classification	ha				
	Crown / Licence	ha				
	Crown Grant	ha				
Brushing and Weeding	Total	ha	107.3	345.1	307.3	759.7
	Manual	ha			66.3	66.3
	Mechanical	ha	107.3	196.9		304.2
	Stem Injection	ha		94.4	223.2	317.6
	Aerial Foliar	ha		10.9	3.3	14.2
	Ground Foliar	ha		42.9	14.5	57.4
	Crown / Licence	ha	107.3	339.9	307.3	754.5
	Crown Grant (MF)	ha		5.2		5.2
Juvenile Spacing	Total	ha	111.4	45.8	160.3	317.5
	Crown / Licence	ha	103.4	45.8	160.3	309.5
	Crown Grant (MF)	ha	8.0			8.0
Pruning	Total	ha	185.3			185.3
	Crown / Licence	ha	167.6			167.6
	Crown Grant (MF)	ha	17.7			17.7
Fertilization	Total	ha	1335.0	1118.2	962.7	3415.9
	Crown / Licence	ha	1335.0	1118.2	962.7	3415.9
	Crown Grant (MF)	ha				
Commercial Thinning	Total	ha				0
	Crown / Licence	ha				
	Crown Grant (MF)	ha				
Residue Assessment Plots			71	157	63	291
CP Cruising Plots			192	431	335	958
Minor Products	Cedar Shake and Shingle	m^3	4701.2	4562.1	2179	11442.3
	Yew Bark	kg	107.6			107.6
	Gravel and Sand	m^3	26597.25			26597.25
	Honey	kg	454		545	545
	Cedar Oil	l	1227.0			1227
	Salvage Logs	m^3	1267.6	195.9		1463.5
	Cedar Cants	m^3	16.5			16.5
	Guitar Blocks	m^3				
Engineering	Roads Constructed	km	19.2	42.4	25.5	87.1
	Roads Re-Built	km	5.8	3.7	19.3	28.8
	Roads Maintained	km	251.3	210.7	353.2	815.2
	Roads Deactivated					
	Temporary	km	26.5	1.5	10.9	38.9
	Semi-Permanent	km	28.3	15.5	8.9	52.7
	Permanent	km	6.9	29.1	2.9	38.9
Roadside Treatments	Mechanical Brushing	km		32	6	38
	Chemical Spraying	km			32	32
	Hydro and Dry Seeding	km		23.5	17.7	41.2
Site Stabilization		ha		12.5	3.9	16.4
Salmonid Enhancement Program	Total		1321074	465510	108550	1895134
	Chinook		1045690	45350	42700	1133740
	Coho		272341	369960	65850	708151
	Chum		0	40000		40000
	Steelhead		3043	10200		13243

Appendix XII

Western Forest Products

**Not Satisfactorily Regenerated Balance Sheet
to December 31, 1999**

Item	TFL 6		TFL 19		TFL 25		Total	
Opening Balance (NSR at 1999 01 01)	1601	ha	1188	ha	2116	ha	4905	ha
Debits								
1999 Denudations	1607	ha	588	ha	619	ha	2814	ha
1999 Surveys	44	ha	131	ha	40	ha	215	ha
Credits								
1999 Planting	984	ha	452	ha	358	ha	1794	ha
1999 Surveys (Natural Regeneration)	190	ha	0	ha	48	ha	238	
1999 Other	0	ha	0	ha	0	ha	0	ha
Closing Balance (NSR at 1999 12 31)	2078	ha	1455	ha	2369	ha	5902	ha

Appendix XIII

TREE FARM LICENCE 6
Historical Summary of Activities

Year	Denuded (ha)	Planted (ha)	No. Trees Planted	Juvenile Spacing (ha)	Brushing (ha)	Prescribed Burning (ha)	Mechanical Site Prep. (ha)	Fertilization (ha)	Pruning (ha)	Commercial Thinning (ha)
Pre 1965	13793.2	797.7	701400	0.0	174.2	484.9	51.4	0.0	0.0	0.0
1965	1517.5	497.2	361500	20.3	0.0	1514.8	0.0	0.0	0.0	0.0
1966	1756.5	430.9	325300	30.4	0.0	1382.0	0.0	0.0	0.0	0.0
1967	1744.8	674.9	422950	16.2	12.2	1654.2	57.0	0.0	0.0	0.0
1968	1780.8	620.0	444900	0.2	54.6	1363.8	0.0	0.0	0.0	0.0
1969	1742.7	1126.1	989650	0.8	0.0	1195.6	0.0	0.0	0.0	0.0
1970	1469.4	1155.0	751700	17.1	0.0	1177.9	14.6	0.0	0.0	0.0
1971	1751.7	717.8	529350	83.0	30.0	593.2	5.7	0.0	0.0	0.0
1972	1310.2	1066.2	912650	136.1	0.0	1175.1	24.7	0.0	0.0	0.0
1973	1660.7	569.6	600500	105.7	6.1	790.0	8.5	0.0	0.0	0.0
1974	1496.5	459.1	459350	37.9	112.6	646.5	5.7	0.0	0.0	0.0
1975	944.5	911.5	777700	2.2	276.2	10.9	4.5	0.0	0.0	0.0
1976	1875.1	983.6	777050	557.7	390.8	1439.6	0.0	0.0	0.0	0.0
1977	2137.4	1047.8	553900	729.3	726.6	1423.8	0.0	3.5	0.0	0.0
1978	1977.2	692.3	493950	720.4	72.9	23.1	0.0	0.0	0.0	0.0
1979	1879.8	837.4	662850	566.5	127.9	1090.0	1.6	0.0	0.0	0.0
1980	1686.4	566.8	491500	462.0	342.3	374.0	0.0	0.0	0.0	0.0
1981	1378.4	1213.7	1047600	333.4	279.8	231.1	0.0	0.0	0.0	40.9
1982	1659.2	1256.7	1198300	172.2	0.0	335.4	0.0	0.0	0.0	0.0
1983	2313.3	936.6	888000	0.0	373.7	571.7	45.3	0.0	0.0	0.0
1984	2315.2	960.2	882400	132.7	1768.9	421.0	24.0	0.0	26.9	0.0
1985	2234.1	648.2	701800	485.6	712.1	1050.8	118.4	0.0	17.3	0.0
1986	1772.1	1301.1	1347100	303.3	1099.9	1526.2	59.2	200.0	9.0	0.0
1987	2284.8	2258.4	2256650	419.6	2668.4	967.4	115.6	0.0	9.1	0.0
1988	1849.5	1866.9	1844050	65.4	899.7	1008.9	11.1	0.0	2.0	0.0
1989	1173.4	1118.8	1169250	469.9	2053.4	874.7	64.7	838.0	5.5	0.0
1990	1394.3	1382.0	1405700	548.0	1511.7	617.0	35.2	795.0	8.4	0.0
1991	1642.6	1801.9	1491100	827.1	1656.2	347.5	50.2	0.0	63.8	0.0
1992	1467.2	1522.5	1550900	377.1	1429.2	491.7	14.2	0.0	34.9	0.0
1993	1881.7	1524.1	1574650	97.5	1357.5	499.7	86.9	0.0	42.4	0.0
1994	1666.8	1697.2	1712150	829.1	843.9	424.2	92.4	0.0	178.5	0.0
1995	1661.7	2060.6	2003400	802.6	1487.1	76.6	47.7	516.1	350.4	7.5
1996	1697.5	2065.8	2110950	549.7	1475.7	267.8	87.8	1936.2	659.9	0.0
1997	1404.2	1789.4	1944750	539.5	1066.3	175.3	24.3	2323.3	916.5	0.0
1998	1104.7	1333.7	1473600	76.4	1074.0	0.0	25.6	0.0	256.6	7.1
1999	1536.3	983.2	1088000	317.5	759.7	0.0	25.4	3415.9	185.3	0.0
TOTAL	72961.4	40874.9	37946550	10832.4	24843.6	26226.4	1101.7	10028.0	2766.5	55.5

Historical Summary amended to include all of the former TFL 25 Block 4

Appendix XIV

Western Forest Products
Tree Planting History

Number of Seedlings

Year	TFL 6	TFL 19	TFL 25	WFP Misc. Properties	Other Properties	TOTAL
Pre 1965	701400	3502000	2531100	3782450		10516950
1965	361500	425000	298500	247700		7996200
1966	325300	726000	432800	0		8904600
1967	422950	434000	547650	285800		10142400
1968	444900	539000	645250	46700		10055100
1969	989650	474000	446100	327300		13422300
1970	751700	535000	341450	136500		10587900
1971	529350	1123000	586700	158250		14383800
1972	912650	912000	295300	407000		15161700
1973	600500	699000	772450	162000		13403700
1974	459350	1324000	363850	57950		13230900
1975	777700	942000	199450	67950		11922600
1976	777050	709000	807250	58250		14109300
1977	553900	631000	757550	172350		12688800
1978	493950	494000	555600	38800		9494100
1979	662850	524000	749000	12300		11688900
1980	491500	473000	493650	24150		8893800
1981	1047600	579000	803900	29900		14762400
1982	1198300	735000	827700	16900		16667400
1983	888000	566000	669050	55450		13071000
1984	882400	325000	809000	102700		12714600
1985	701800	452000	522050	69550		10472400
1986	1347100	346000	630950	57000		14286300
1987	2256650	686000	1297750	329300		27418200
1988	1844050	563000	982850	172950		21377100
1989	1169250	755000	735600	287750		17685600
1990	1405700	707000	712350	354150		19075200
1991	1491100	439000	842850	60150		16998600
Pre 1992					5268800	31612800
1992	1550900	757000	673900	95700	232850	19862100
1993	1574650	683000	639750	240600	377550	21093300
1994	1712150	674000	546000	226200	790600	23693700
1995	2003400	1040000	853050	160400	894500	29708100
1996	2110950	1140000	1089750	579400	3550	29541900
1997	1944750	1067000	951900	52150	0	24094800
1998	1473600	675550	652250	1522400	28750	4354548
1999	1088000	382881	413000	901550	21900	2809330
TOTAL	13,458,400	6419431	5819600	3778400	7618500	37,094,331

Appendix XV – A

Employment Statistics – TFL 6

Operation	Home Region	Contractor Personnel		Company Personnel		TOTAL	
		People	Person Days	People	Person Days	People	Person Days
Planning, Engineering and Road Development							
Head Office	South Mainland	0	0	1	91	1	91
Port McNeill	North Vancouver Island	15	1220	8	1401	23	2621
Jeune Landing	North Vancouver Island	15	2490	6	1143	21	3633
	South Mainland	1	50		0	1	50
Holberg	North Vancouver Island	42	4724	11	1525	53	6249
	Vancouver Island	1	69	3	579	4	648
	South Mainland	1	5	0	0	1	5
SUBTOTAL		75	8558	14	4739	104	13297
Harvesting							
Port McNeill	North Vancouver Island	12	2252	67	16843	79	19095
	Vancouver Island	17	2822	0	0	17	2822
Quatsino Dryland Sort	North Vancouver Island	90	9232	0	0	90	9232
	Vancouver Island	3	213	0	0	3	213
Jeune Landing	North Vancouver Island	0	0	9	1252	9	1252
	Vancouver Island	76	1323	0	0	76	1323
	South Mainland	29	358	0	0	29	358
Holberg	North Vancouver Island	81	15317	59	8098	140	23415
	Vancouver Island	12	2140	7	739	19	2879
	South Mainland	2	159	2	213	4	372
SUBTOTAL		322	33816	144	27145	466	60961
Silviculture and Integrated Resource Management							
Head Office	South Mainland	12	333	5	356	17	689
Saanich	Vancouver Island	10	241	55	2096	65	2337
Port McNeill	North Vancouver Island	0	0	10	1172	10	1172
	Vancouver Island	71	3681	0	0	71	3681
Jeune Landing	North Vancouver Island	0	0	9	1252	9	1252
	Vancouver Island	76	1323	0	0	76	1323
	South Mainland	29	358	0	0	29	358
Holberg	North Vancouver Island	4	82	12	1398	16	1480
	Vancouver Island	66	871	0	0	66	871
	South Mainland	35	720	0	0	35	720
SUBTOTAL		303	7609	91	6274	394	13883

Operation	Home Region	Contractor Personnel		Company Personnel		TOTAL	
		People	Person Days	People	Person Days	People	Person Days
Transportation							
Head Office	South Mainland	79	6671	5	364	56	6981
SUBTOTAL		79	6671	5	364	56	6981
Processing							
Port Alice Pulp Mill	Vancouver Island			451	55216	451	55216
Squamish Pulp Mill	Lower Mainland			344	24296	344	24296
Ladysmith Sawmill	Vancouver Island			90	14176	90	14176
Cowichan Bay Sawmill	Vancouver Island			95	2456	95	2456
Silvertree Sawmill	Lower Mainland			118	6770	118	6770
Tahsis Sawmill	Vancouver Island			200	3992	200	3992
Duke Point Sawmill	Vancouver Island			80	18240	80	18240
Nanaimo Sawmill	Vancouver Island			152	1093	152	1093
Log Trading and Sales	Lower Mainland			40	2699	40	2699
Nanaimo Log Merchandising	Vancouver Island			36	880	36	880
SUBTOTAL				1606	129818	1606	129818
Administration							
Head Office	South Mainland	4	116	31	2825	35	2941
Region	Vancouver Island			6	1320	6	1320
SUBTOTAL		4	116	37	4145	41	4261
Summary – By Home Region							
	North Vancouver Island	259	35317	191	34084	450	69401
	Vancouver Island (Other)	332	12683	1175	100787	1507	113470
	South/Lower Mainland	192	8770	546	37614	710	46330
TOTAL		192	8770	546	37614	710	46330

Appendix XV - B

Western Forest Products
Direct Employment Statistics*

(person-days)

	Planning and Development	Harvesting	Transportation	Processing	Silviculture and Integrated Resources Management	Administration	TOTAL
TFL 6	13297	60961	6981	129818	13883	4261	**229201**
TFL 19	12693	38491	4602	62113	7754	2423	**128076**
TFL 25	5765	21262	6082	43376	7294	1431	**85210**
MF 61	456	1310	23	**	36	121	**1946**
FL A16845	1124	7749	847	11675	890	843	**23128**
FL A16847	1148	10599	655	28332	1665	936	**43335**
FL A19205	28	**	63	4812	418	125	**5446**
FL A19216	33	**	90	9992	367	171	**10653**
FL A19221	8	**	**	**	66	13	**87**
FL A19228	203	**	91	6802	400	159	**7655**
FL A19231	3938	25949	3420	46321	1884	1907	**83419**
FL A19240	586	2251	253	3845	245	122	**7302**
FL A53746	70	1725	**	**	295	25	**2115**
Other Tenures	378	1507	170	27450	939	358	**30802**
TOTAL	**39727**	**171804**	**23277**	**374536**	**36136**	**12895**	**658375**

***** Includes Company and Contract Personnel.
****** See Other Tenures

Appendix XV - C

Western Forest Products
First Nations Silviculture Contracts Employment Summary
1999

TENURE	OPERATION							
	Holberg	Jeune Landing	Port McNeill	Mainland Islands Region	Zeballos	Gold River	Nootka Contract Admin	TOTAL
TFL 6	52	0	2105					2157
TFL 19					886	969	99	1954
TFL 25				1668				1668
MF 61								
FL A16845				108				108
FL A16847				180				180
FL A19205				46				46
FL A19216				12				12
FL A19221								
FL A19228								
FL A19231					243		80	80
FL A19240								
FL A53746								
Other Tenures								
River's Inlet				147				147
Mathieson Channel				292				292
Campbell Island				74				74
TOTAL	52	0	2105	2453	1129	969	179	6887
TOTAL Contractor Days	1673	1681	3681	6257	2775	1483	2049	19599
% FN Employment	3	0	!Zero Divide	!Zero Divide	!Zero Divide	!Zero Divide	!Zero Divide	35
% WFP Goal	15	0	0	0	0	0	0	175

Appendix XVI

Tree Farm Licence 6
Salmonid Enhancement Program Summary

No. Fry Released (000)

Year	Hatchery	Release Stream	Coho	Chum	Chinook*	Steelhead	Pink	Subtotal	TOTAL
1999	Marble	Marble River	272.4	0	1045.7	3.0		1321.1	
	Colonial	Colonial Creek	65.9	0	42.7			108.6	
	Cordy	Goodspeed River	222.9	40.0	45.4	10.2		318.5	
		San Josef River	146.9					146.9	1895.1
1998	Marble	Marble River	37.2	0	201.4	3.2		241.8	
	Colonial	Colonial Creek	60.5		7.0			67.5	
	Cordy	Goodspeed River	23.9	20.6	3.5			48.0	
		San Josef River	42.4					42.4	399.7
1997	Marble	Marble River	44.7		741.3			1,321.1	
	Colonial	Colonial Creek	19.7		56.7	9.7		108.6	
	Cordy	Goodspeed River	10.9	65.4	29.1			318.5	
		Holberg Inlet			18.9			146.9	
		San Josef River	49.2					241.8	1045.6
1996	Marble	Marble River	110.7		127.2			237.9	
	Colonial	Colonial Creek	65.8		12.8			78.6	
	Cordy	Goodspeed River	32.9	33.6	19.6			86.1	
		San Josef River	50.4					50.4	
		Pegattem River		9.3				9.3	462.3
1995	Marble	Marble River	40.0		300.0	6.0		346.0	
	Colonial	Colonial Creek	80.1		38.2			118.3	
	Cordy	Goodspeed River*	95.0	141.9	17.1	0.4		254.4	
		San Josef River	8.3					237.9	727.0
1994	Marble	Marble River	42.0		700.0	6.0		78.6	
	Colonial	Colonial Creek*	80.0		36.8	10.9		127.7	
	Cordy	Goodspeed River*	13.9	34.4	21.0	7.5		76.8	
		San Josef River	31.0					31.0	983.5
1993	Marble	Marble River	42.0		74.0	8.0		346.0	
	Colonial	Colonial Creek	19.5					118.3	
	Cordy	Goodspeed River	20.1	62.0		7.8		254.4	
		San Josef River	33.0					8.3	266.4
1992	Marble	Marble River	78.0		312.2			748.0	
	Colonial	Colonial Creek*	109.7		19.1			128.8	
	Cordy	Goodspeed River*	98.7	82.8	21.5	7.0	2.2	76.8	
		San Josef River	1.0					31.0	732.2
1991	Marble	Marble River	49.3		65.0	7.0		124.0	
	Colonial	Colonial Creek	100.0					19.5100.0	
	Cordy	Goodspeed River	76.2	42.4		7.7	11.0	89.9	
		San Josef River	25.4					33.0	359.6

Year	Hatchery	Release Stream	Coho	Chum	Chinook*	Steelhead	Pink	Subtotal	TOTAL
1990	Marble	Marble River	54.3		629.8	9.0		638.8	
	Colonial	Colonial Creek*	44.0		26.5	14.4		84.9	
	Cordy	Goodspeed River*	92.4	77.0	30.0			199.4	977.4
1989	Marble	Marble River	24.0	0	120.0	1.4		145.4	
	Colonial	Colonial Creek*	27.0	0	6.3			33.3	
	Cordy	Goodspeed River*	27.0	65.0	30.5	9.0		131.5	319.2
1988	Marble	Marble River	49.6		350.0			399.6	
		Waukwass				7.7		7.7	
	Colonial	Colonial Creek*	80.0		8.0			88.0	
	Cordy	Goodspeed River*	15.5	102.0	8.8	3.3		129.6	
	Monkey	Mahatta River	95.0					95.0	
		Monkey Creek	9.5	19.0				28.5	748.4
to 1987	Marble	Marble River	122.0	2.8	1,243.7			1368.5	
	Cordy	Goodspeed	132.0	71.0				203.0	
	Colonial	Colonial Creek	69.0			3.0		72.0	
	Nequiltpaalis	Nequiltpaalis		42.0				42.0	
	Monkey	Mahatta River	123.0					123.0	
		Monkey Creek	33.0	87.0				120.0	1928.5
TOTAL									**10844.9**

Appendix XVII

WESTERN FOREST – WESTERN PULP – DOMAN INDUSTRIES – DOMAN WESTERN LUMBER

1999 Log Flow and Wood Consumption
(Approximate)

SAWMILL PULPMILL	TENURE / SOURCE (THOUSAND CUBIC METRES)													Chips to Woodfibre (Thousand Units)
	TFL 6	TFL 19	TFL 25	FL A19240 Strathcona	FL A19240 Kingcome	FL A16845 Mid Coast	FL A16847 Mid Coast	FL A19205 Fraser	FL A19228 Sunshine	FL A19216 Soo	Other Tenures	Inventory / Purchase	Total Consumption	
Duke Point	460				14						3	-183	294	45
Chemainus											1	113	114	
Ladysmith	237		54		5	14		2	4	1	6	30	353	0
Cowichan	33	31	53	32	2	9	5	2	17	2	6	56	248	
Silvertree	143	61	58	91	7	25	19				10	-121	293	44
Vancouver		28	34	20		25		9		7	14	56	193	28
Tahsis	50	165		117							13	-17	328	48
Saltair			34				24			18	5	426	507	
Nanaimo	5	30	15	10			60	2	4	6		346	478	0
Log Merchandizer	16	17	33	12		16	2	12	21	7	8	630	774	360
Port Alice Pulp Mill	263	81	17	57	8		23				5	114	568	
Trades / Sales	169	130	63	100	9	32	22	4	1	4	17	76	627	(3)
Squamish Pulp Mill Chips														
• Purchased														222
• Consumed														734
TOTAL LOGS	1376	543	361	439	45	121	155	31	47	45	88	1526	4777	

Appendix XVIII

Western Forest Products Limited
Forest Research Summary
Permanent Plots and Trials

Trial *(Year Established)*	Location	Measurements [1]	Reports	Other
Forest Nutrition and Salal Cedar Hemlock Integrated Research Program				
SCHIRP Establishment (88)	Port McNeill	97, 94, 90, 88, 87	96 Update	Foliar Sampling (97)
Sitka Spruce Demonstration (84)	Port McNeill	98, 96, 90, 89, 88, 87, 86, 85, 84	94	Foliar Sampling (97) Re-fertilize (97)
Western Red Cedar Demonstration (87)	Port McNeill	98, 96, 94, 93, 92, 91, 90, 89, 88, 87, 86	94	Foliar Sampling (97) Re-fertilize (97)
Western Hemlock Demonstration (87)	Port McNeill	98, 96, 94, 93, 92, 91, 90, 89, 88, 87, 86	94	Foliar Sampling (97) Re-fertilize (97)
Salal Eradication (84)	Port McNeill	94	96 Update	
S1CH Scarification (96)	Port McNeill	97, 95	96 Est. Rep.	Salal Measures (98)
S1CH Individual Tree Fertilization (96)	Port McNeill	97, 96, 95	96 Est. Rep.	
Individual Tree Fertilization (96)	Holberg	97, 95	96 Est. Rep.	
Transitional S1CH/S1HA Fertilization (96)	Holberg Port McNeill	97, 95	96 Est. Rep.	Salal Measures (98)
Salal Fertilization (93)	Port McNeill	93	94	
Operational Hand Fertilization (96)	Port McNeill Holberg	97,95	96 Est. Rep.	
Aerial Fertilization (97)	Port McNeill Holberg	96		
Aerial Fertilization (96)	Port McNeill	97, 95		
Aerial Fertilization (95)	Port McNeill	99, 96, 94		
Aerial Fertilization (90)	Port McNeill Holberg Jeune Landing	95, 90, 89	97 Interim Rep.	
Aerial Fertilization (89)	Port McNeill Holberg	94, 90, 89	97 Interim Rep.	
Aerial Fertilization (86)	Port McNeill	91, 87, 86	94	
Organic Fertilizer (90)	Port McNeill	95, 93, 91	96 Update	With UBC
Fish Silage (93)	Port McNeill	94, 93	96 Update	With UBC
Straw and Fish Compost (94)	Port McNeill	95, 94, 93, 92	96 Update	With UBC
Fertilization by Stock Type (98)	Port McNeill	99, 98, 97	98 Est. Rep.	
Vaccinium SCHIRP (99)	Holberg	99		With Pacific Forestry Centre

Trial *(Year Established)*	Location	Measurements [1]	Reports	Other
		Genetics Trials		
Yellow Cypress Pilot (88)	Port McNeill	96, 95, 94, 93, 91, 90, 89, 88		
Yellow Cypress 1 (91)	Port McNeill, Jeune Landing	97, 94	96 Interim Rep.	
Yellow Cypress 2 (92)	Port McNeill	98, 95		
Yellow Cypress 3 (93)	Port McNeill, Jeune Landing	99, 96		
Yellow Cypress 4 (93)	Port McNeill	99, 96		
Yellow Cypress 5 (94)	Port McNeill	97		
Yellow Cypress 6 (95)	Port McNeill	98		
Yellow Cypress 7 (96)	Holberg	99		
Yellow Cypress 8 (97)	Port McNeill			
Yellow Cypress 9 (99)	Port McNeill			
Coastal Douglas-fir Progeny Trial (93)	Port McNeill	97	93 Est. Rep.	
Hybrid Poplar Clonal Trial (91)	Holberg	93, 91	91 Est. Rep.	
Sitka Spruce Clonal Trial (88)	Holberg		88 Est. Rep.	
		Other Silviculture Research		
Suquash Drainage (97)	Port McNeill	99, 98, 97	98 Est. Rep.	
Western Red Cedar Stock Type (96)	Port McNeill	97, 96	97 Est. Rep	
S4 Sitka Spruce (84)	Holberg	95, 90, 88, 87, 86, 85, 84	89 5-Year Results	2nd Pruning and Assessments
Cottonwood Nurse Tree (94)	Jeune Landing		96 Est. Rep.	
Weevil Resistant Sitka Spruce (97)	Jeune Landing		97 Est. Rep.	
		Growth and Yield Monitoring		
Site Index Species Conversion Surveys (97)	Port McNeill Holberg Jeune Landing	97		
Type III Growth and Yield Installations (88)	Port McNeill Holberg	95	95 MOF Rep.	Vegetation Measures (97)
Growth and Yield (94)	Port McNeill	94		
Growth and Yield (93)	Port McNeill	93		
Growth and Yield (92)	Port McNeill	92		
Growth and Yield (91)	Port McNeill Jeune Landing	91		
Growth and Yield (89)	Port McNeill Jeune Landing	89		
Growth and Yield (88)	Port McNeill Holberg Jeune Landing	88		

Appendix XIX

Doman Forest Products Limited
Western Pulp Inc.
Western Forest Products Limited
Doman – Western Lumber Ltd.

Operating Statistics

Productive Forest Land Managed	885 000 ha
Operable Forest Land	550 000 ha
Forest Tenures	3 Tree Farm Licences
	7 Forest Licences
	5 Managed Forests
	127 Timber Licences
Logging Operations	30
Employees and Contractors	4 200 people (est)
Annual Timber Harvest	4 200 000 m^3
Annual Timber Purchase	800 000 m^3
Mills	2 Pulp Mills – 1 Kraft, 1 Sulphite
	9 Saw Mills
	1 Value Added Plant
	1 Log Merchandiser
Products: Lumber and Solid Wood	800 000 000 board feet
Pulp	400 000 tonnes
Annual Product Sales	over $800 million
Annual Roads Construction	335 km
Annual Roads Maintenance	2 100 km
Annual Logging	5 500 ha
Annual Planting	5 100 ha
Annual Natural Regeneration	400 ha
Annual Number of Seedlings Planted	5 000 000 trees
Average Survival of 3-year Old Plantations	90 %
Annual Site Preparation	700 ha
Annual Brushing and Weeding	2 300 ha
Annual Juvenile Spacing	1 600 ha
Annual Pruning	1 000 ha
Annual Fertilization	3 000 ha
Annual visitors to Forest Lands	over 300 000 visitors
Annual Salmon Enhancement Production (4 hatcheries)	750 000 fry
Recreation Sites and Trails	45
Forest Enhancement Person-Days	over 40 000

Appendix XX

Summary of Obligations
Management Plan Commitments and Issues*
Status Report

Commitment	Timeline	Status
Refine Site Index/Ecosite Relationship	Prior to MP 9 draft	A project was completed by J.S.Thrower & Associates that confirmed relationship.
Confirm yield curve and stand density relationship	May 31, 1996 for review. Next MP for changes.	A project has been undertaken with Olympic Resources to utilize PSP data to project inventory data through time. The results will be available in 1999.
Address Inventory gaps	Prior to preparation of 20 year plan (Dec 96 Target).	**Recreation** – Opportunity mapping done and inputted into GIS. **Terrain/Silviculture** – Completed and inputted into GIS. **Wildlife** – Habitat prediction models were developed and tested in 1999. Models were found to yield accurate results. **Stream Classification** – RIC inventory in progress to supplement operational work. Archaeological Sites – Captured from HCB records. **Biodiversity** – T. Lewis completed ecosystem and forest cover work to supplement D. Blood report. **TRP** – Spatial mapping tool was switched to COMPLAN for use in MP9
Deciduous stands will be harvested and replaced with conifers through the rotation as opportunities arise.	No date.	Harvesting of deciduous stands was included as part of the long term harvesting plan in MP 8. However, such stand conversion was viewed as occurring concurrently with the harvest of adjacent second growth coniferous stands. WFP has not yet reached this point in time and very little harvesting is occurring in these younger stands. We have invited deciduous contractors to bid on stand conversion.
Accounting for non-recoverable losses.	During term of MP.	Pilot windthrow project is completed. Data being analysed for expanded program.
Feasibility study of commercial thinning	During term of MP.	A series of CT trials have been completed using different systems (cable, CTL and hoe-forwarding. Results are not encouraging for both economic and forest health reasons. Projects have been uneconomic and level of damage in hemlock stands has been unacceptable.

*Outstanding commitments and issues that were contained in the implementation section of the TFL 6 AAC rationale, the approval letter and in the AAC rationale report for management Plan 8.

Appendix XXI

Saanich Forestry Centre
Seedling Production Report

	Seedlings Produced					
Species	**Size**	**Fall**	**Spring**	**Total by Size**	**Total by Species**	**%**
Cw	313B	12,800	423,710	436,510		
Cw	410A	0	215,450	215,450		
Cw	415C	16,620	524,750	541,370		
Cw	615A	80	17,030	17,110		
Total					1,210,440	49.58
Dr	313B	0	3,780	3,780		
Dr	415B	0	1,770	1,770		
Total					5,550	0.23
Ds	313B	0	4,120	4,120		
Ds	415B	1,128	1,060	2,188		
Total					6,308	0.26
Fc	410A	0	83,760	83,760		
Fc	415C	0	443,280	443,280		
Fc	615A	603	5,930	6,533		
Total					533,573	21.85
Hm	410A	1,850	1,440	3,290		
Hm	415C	0	10,380	10,380		
Total					13,670	0.56
Hw	313B	0	90,880	90,880		
Hw	410A	0	71,280	71,280		
Hw	415C	18,890	390,440	409,330		
Hw	615A	0	9,750	9,750		
Total					581,240	23.81
Plc	313B	50	4,480	4,530		
Plc	410A	0	2,690	2,690		
Total					7,220	0.30
Pw	410A	0	4,240	4,240		
Pw	415C	0	12,640	12,640		
Total					16,880	0.69
Ss	313B	0	2,800	2,800		
Ss	415C	240	19,010	19,250		
Total					22,050	0.90
Sx	415C	21,180	21,270	42,450		
Total					42,450	1.74
Yc	410A	1,650	0	1,650		
Total					1,650	0.07
Misc	415C	500	0	500		
Misc	615A	90	0	90		
Total					590	0.02
Total		75,681	2,365,940	2,441,621	2,441,621	100.00

Appendix XXII

Saanich Forestry Centre
Seed Production Report

Seedlot	Species	Orchard	Zone	BVvol60[1] (%)	Elevation (m)	Latitude	Longitude	Volume (hL)	Seed Weight (Kg)	Seedlings (000s)	Comments
61023	Fdc	166	M/GL	10	359	49°14'	124°11'	41.82	23.632	716.3	SMP Crop
61024	Fdc	169	M/GL	5	259	49°05'	123°52'	5.14	2.214	86.8	
61030	Hw	127	M	2	736	50°33'	127°18'	6.64	5.960	1,236.9	
61031	Hw	126	M	8	85	50°34'	127°02'	3.55	2.457	755.4	
61032	Hw	126	M	5	110	50°35'	127°11'	4.98	4.861	933.6	
61025[2]	Cw	155	M	10	241	52°41'	131°37'	3.56	1.708	374.2	SMP Crop
61026[2]	Cw	155	M	5	244	52°42'	131°38'	9.83	6.502	1,653.8	
61027	Cw	155	M	2	239	52°41'	131°38'	31.71	21.363	5,056.2	
61028[2]	Cw	128	M	10	164	50°45'	127°34'	10.64	5.666	1,479.4	SMP Crop
61029	Cw	128	M	2	162	50°46'	127°32'	33.87	20.667	4,877.4	
60699	Cw	155	M	2	345	52°46'	131°45'	3.10	2.062	480.9	
								Total:	**97.092**	**17,650.9**	

1 Expected gain in volume over wild seedlots at age 60.
2 BV vol. 60 estimated, awaiting confirmation from DNA out-crossing studies.